॥वसुधैव कुटुम्बकम्॥
SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Practical No.: 11

Aim –

Demonstrate JavaScript timing events (setInterval, setTimeout, clearInterval) and ES6 modular code structure (import/export) by developing an interactive Countdown Timer and Class Scheduler application. Implement Case Study 1: TechNova Technical Workshop Ticket Booking System integrating multi-category registration, dynamic discount calculations, validation, and real-time order summaries, alongside interactive web portals for Smart India Hackathon (SIH) and Rakhi World e-commerce showcase.

Software/Tools Required –

VSCode
Chrome Browser
HTML5
CSS3
JavaScript (ES6 Modules)

Experiment Program Code – Countdown Timer & Class Scheduler

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Countdown Timer & Class Scheduler</title>

    <link rel="stylesheet" href="style.css">
</head>
<body>

    <div class="container">

        <h1>Countdown Timer</h1>
        <p class="subtitle">Manage your study time efficiently!</p>

        <!-- Countdown Timer -->
        <section class="card">
            <h2>Study Timer</h2>
```

```
            <label for="minutes">Enter Minutes:</label>
            <input type="number" id="minutes" min="1" value="5">

            <div id="countdown">05:00</div>

            <p id="status">Ready to start?</p>

            <div class="buttons">
                <button id="startBtn">Start</button>
                <button id="pauseBtn">Pause</button>
                <button id="resetBtn">Reset</button>
            </div>
        </section>

        <!-- Class Scheduler -->
        <section class="card">
            <h2>Class Scheduler</h2>

            <form id="scheduleForm">

                <label for="className">Class Name:</label>
                <input
                    type="text"
                    id="className"
                    placeholder="Enter class name"
                    required
                >

                <label for="classTime">Date and Time:</label>
                <input type="datetime-local" id="classTime" required>

                <button type="submit">Add Class</button>

            </form>

            <p id="message"></p>

            <h3>Scheduled Classes</h3>
            <div id="classList">No classes scheduled.</div>
        </section>

    </div>

    <script type="module" src="main.js"></script>

</body>
</html>
```

style.css

```
* {
    box-sizing: border-box;
}

body {
    margin: 0;
    padding: 20px;
    font-family: Arial, sans-serif;
    background: #edf3ff;
    color: #243451;
```

```
}

.container {
    max-width: 700px;
    margin: auto;
}

h1 {
    text-align: center;
    color: #315fc4;
}

.subtitle {
    text-align: center;
    color: #64748b;
    margin-bottom: 25px;
}

.card {
    background: white;
    padding: 25px;
    margin-bottom: 22px;
    border-radius: 15px;
    box-shadow: 0 5px 18px rgba(30, 60, 100, 0.10);
}

h2 {
    margin-top: 0;
    color: #315fc4;
}

label {
    display: block;
    margin-top: 15px;
    margin-bottom: 7px;
    font-weight: bold;
}

input {
    width: 100%;
    padding: 11px;
    border: 1px solid #cbd5e1;
    border-radius: 7px;
    font-size: 15px;
}

input:focus {
    outline: 2px solid #a5bfff;
    border-color: #315fc4;
}

#countdown {
    text-align: center;
    font-size: 65px;
    font-weight: bold;
    color: #315fc4;
    margin: 22px 0 10px;
}

#status,
#message {
```

```
    text-align: center;
    color: #64748b;
    min-height: 20px;
}

.buttons {
    display: flex;
    justify-content: center;
    gap: 10px;
    flex-wrap: wrap;
}

button {
    padding: 11px 20px;
    margin-top: 15px;
    border: none;
    border-radius: 7px;
    background: #315fc4;
    color: white;
    font-size: 15px;
    cursor: pointer;
}

button:hover {
    background: #23489b;
}

#classList {
    margin-top: 12px;
    padding: 12px;
    background: #f5f8ff;
    border-radius: 8px;
    overflow-wrap: anywhere;
}

.class-item {
    padding: 12px 5px;
    border-bottom: 1px solid #dbe4f3;
}

.class-item:last-child {
    border-bottom: none;
}

.class-item strong {
    display: block;
    margin-bottom: 6px;
}

@media (max-width: 480px) {
    body {
        padding: 12px;
    }

    .card {
        padding: 18px;
    }

    #countdown {
        font-size: 48px;
    }
```

```
    button {
        padding: 10px 14px;
    }
}
```

timer.js

```
// Convert seconds into MM:SS format
export function formatTime(seconds) {
    let minutes = Math.floor(seconds / 60);
    let remainingSeconds = seconds % 60;

    return String(minutes).padStart(2, "0") + ":" +
           String(remainingSeconds).padStart(2, "0");
}

// Start the countdown
export function startCountdown(seconds, onTick, onComplete) {

    let remaining = seconds;

    onTick(remaining);

    const intervalId = setInterval(function () {

        remaining--;

        onTick(remaining);

        if (remaining <= 0) {
            clearInterval(intervalId);
            onComplete();
        }

    }, 1000);

    return intervalId;
}

// Demonstration of setTimeout()
export function notifyAfterDelay(callback) {

    setTimeout(function () {
        callback();
    }, 1000);

}
```

main.js

```
import {
    formatTime,
    startCountdown,
    notifyAfterDelay
} from "./timer.js";

const minutesInput = document.getElementById("minutes");
const countdown = document.getElementById("countdown");
const status = document.getElementById("status");
```

```
const startBtn = document.getElementById("startBtn");
const pauseBtn = document.getElementById("pauseBtn");
const resetBtn = document.getElementById("resetBtn");

let remainingSeconds = Number(minutesInput.value) * 60;
let intervalId = null;
let isRunning = false;

// Display the timer
function updateDisplay() {
    countdown.textContent = formatTime(remainingSeconds);
}

// Stop the timer
function stopTimer() {
    if (intervalId !== null) {
        clearInterval(intervalId);
        intervalId = null;
    }

    isRunning = false;
}

// Start button
startBtn.addEventListener("click", function () {

    if (isRunning) {
        return;
    }

    if (remainingSeconds <= 0) {
        const minutes = Number(minutesInput.value);

        if (!Number.isFinite(minutes) || minutes < 1) {
            status.textContent = "Enter at least 1 minute.";
            return;
        }

        remainingSeconds = Math.floor(minutes * 60);
    }

    isRunning = true;
    status.textContent = "Timer started! Focus on your work.";

    intervalId = startCountdown(
        remainingSeconds,

        function (seconds) {
            remainingSeconds = seconds;
            updateDisplay();
        },

        function () {
            remainingSeconds = 0;
            isRunning = false;
            intervalId = null;

            status.textContent = "Time is up! Take a break.";
```

```
            // Use setTimeout() for delayed notification
            notifyAfterDelay(function () {
                alert("Countdown completed!");
            });
        }
    );

});

// Pause button
pauseBtn.addEventListener("click", function () {

    if (isRunning) {
        stopTimer();
        status.textContent = "Timer paused.";
    }

});

// Reset button
resetBtn.addEventListener("click", function () {

    stopTimer();

    const minutes = Number(minutesInput.value);

    if (Number.isFinite(minutes) && minutes >= 1) {
        remainingSeconds = Math.floor(minutes * 60);
    } else {
        remainingSeconds = 300;
    }

    updateDisplay();
    status.textContent = "Timer reset.";

});

// Update timer when duration changes
minutesInput.addEventListener("change", function () {

    if (isRunning) {
        return;
    }

    const minutes = Number(minutesInput.value);

    if (Number.isFinite(minutes) && minutes >= 1) {
        remainingSeconds = Math.floor(minutes * 60);
        updateDisplay();
        status.textContent = "Ready to start?";
    } else {
        status.textContent = "Enter at least 1 minute.";
    }

});



// ---------------- CLASS SCHEDULER ----------------
```

```
const scheduleForm = document.getElementById("scheduleForm");
const classNameInput = document.getElementById("className");
const classTimeInput = document.getElementById("classTime");

const classList = document.getElementById("classList");
const message = document.getElementById("message");

// Load saved classes from localStorage
let classes = JSON.parse(
    localStorage.getItem("scheduledClasses") || "[]"
);

// Display scheduled classes
function displayClasses() {

    classList.replaceChildren();

    if (classes.length === 0) {
        classList.textContent = "No classes scheduled.";
        return;
    }

    classes.sort(function (a, b) {
        return new Date(a.time) - new Date(b.time);
    });

    classes.forEach(function (item) {

        const div = document.createElement("div");
        div.className = "class-item";

        const title = document.createElement("strong");
        title.textContent = item.name;

        const time = document.createElement("p");
        time.textContent = new Date(item.time).toLocaleString();

        div.appendChild(title);
        div.appendChild(time);

        classList.appendChild(div);

    });

}

// Add a new class
scheduleForm.addEventListener("submit", function (event) {

    event.preventDefault();

    const name = classNameInput.value.trim();
    const time = classTimeInput.value;

    if (!name || !time) {
        message.textContent = "Please fill in all fields.";
        return;
    }
```

```
    if (new Date(time).getTime() <= Date.now()) {
        message.textContent = "Please select a future date and time.";
        return;
    }

    classes.push({
        name: name,
        time: time
    });

    // Save classes in browser storage
    localStorage.setItem(
        "scheduledClasses",
        JSON.stringify(classes)
    );

    displayClasses();

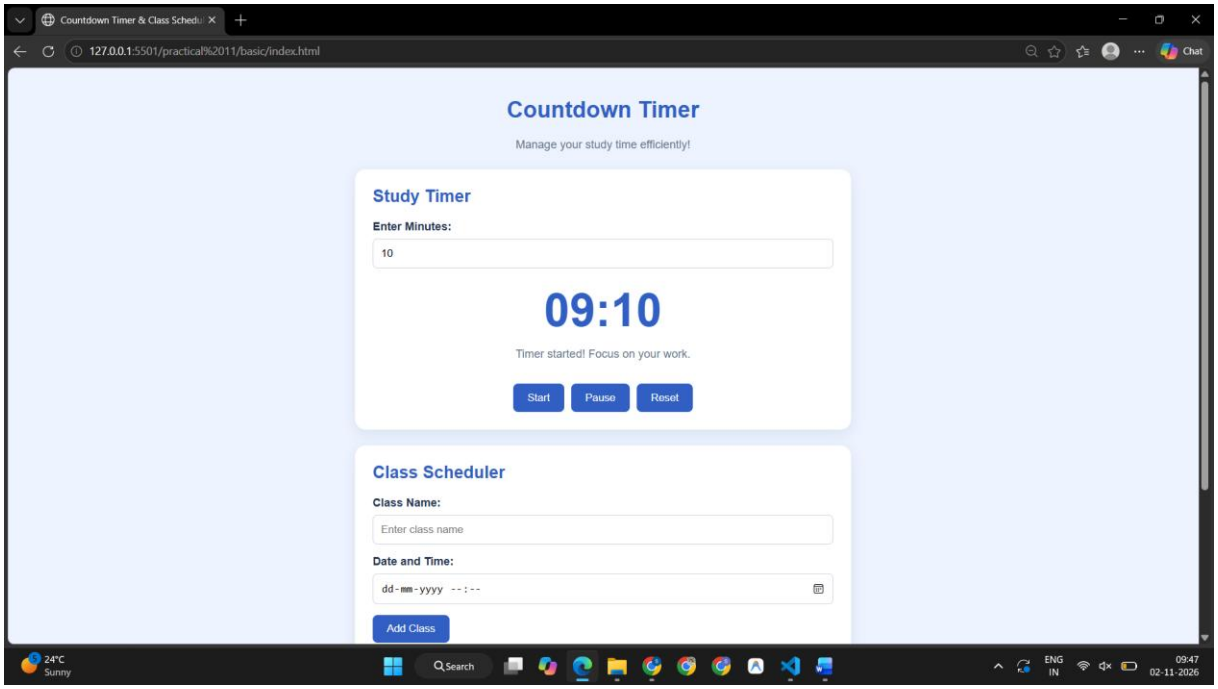
    message.textContent = "Class added successfully!";

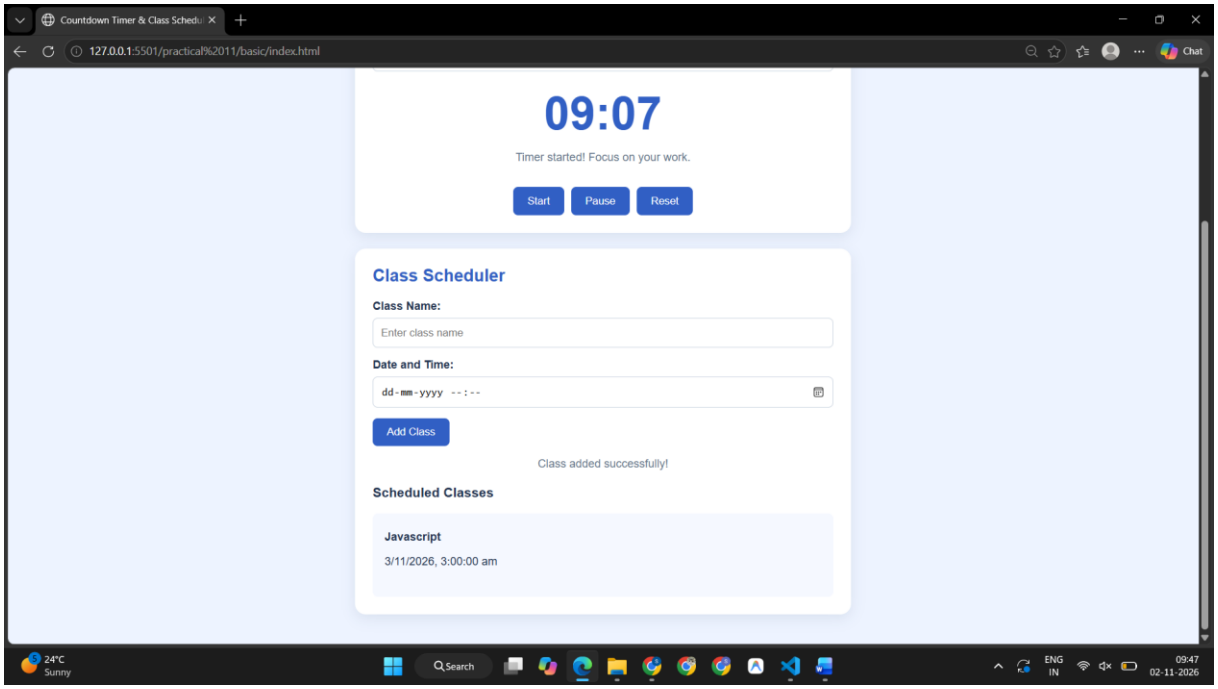
    scheduleForm.reset();

});

// Initial display
updateDisplay();
displayClasses();
```

Output –





Case Study Title –

Case Study 1: TechNova Technical Workshop Booking System with Dynamic Ticket Calculation and Validation

Case Study Program Code –

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>TechNova | Workshop Ticket Booking</title>

    <!-- External CSS -->
    <link rel="stylesheet" href="style.css">

    <!-- Internal JavaScript: small page-level welcome action -->
    <script>
        console.log("Technical Workshop Booking page is loading...");

        function showWelcomeMessage() {
            console.log("Welcome to TechNova Technical Workshop!");
        }

        showWelcomeMessage();
    </script>
</head>

<body>
```

```
<!-- Header -->
<header class="header">
    <div class="logo">
        <span class="logo-icon">⚡</span>
        <div>
            <h2>TechNova</h2>
            <small>Learn • Build • Innovate</small>
        </div>
    </div>

    <nav>
        <a href="#workshop">Workshop</a>
        <a href="#booking">Book Ticket</a>
        <a href="#summary">Summary</a>

        <button id="themeToggle" class="theme-btn" type="button">
            🌙
        </button>
    </nav>
</header>



<!-- Hero Section -->
<section class="hero" id="workshop">

    <div class="hero-content">

        <span class="badge">🚀 TECHNICAL WORKSHOP 2026</span>

        <h1>
            Build Your Future
            <span>With Technology.</span>
        </h1>

        <p>
            Join industry experts for a hands-on technical workshop
            covering AI, Web Development, Cloud and emerging technologies.
        </p>

        <div class="hero-info">

            <div>
                <span>📅</span>
                <div>
                    <small>Date</small>
                    <strong>18 October 2026</strong>
                </div>
            </div>

            <div>
                <span>📍</span>
                <div>
                    <small>Venue</small>
                    <strong>TechNova Auditorium</strong>
                </div>
            </div>

            <div>
                <span>⏰</span>
```

```
                <div>
                    <small>Duration</small>
                    <strong>10:00 AM – 5:00 PM</strong>
                </div>
            </div>

        </div>

        <a href="#booking" class="primary-btn">
            Book Your Ticket →
        </a>

    </div>

    <div class="hero-visual">
        <div class="floating-card card-one">
            <span>🤖</span>
            <div>
                <strong>AI & ML</strong>
                <small>Hands-on Session</small>
            </div>
        </div>

        <div class="tech-circle">
            <div class="circle-inner">
                <span>💻</span>
                <strong>TECH</strong>
                <small>WORKSHOP</small>
            </div>
        </div>

        <div class="floating-card card-two">
            <span>☁️</span>
            <div>
                <strong>Cloud</strong>
                <small>Industry Experts</small>
            </div>
        </div>
    </div>

</section>



<!-- Workshop Highlights -->
<section class="highlights">

    <div class="section-heading">
        <span class="mini-label">WHAT YOU'LL GET</span>
        <h2>More Than Just a Workshop</h2>
        <p>
            Everything you need for a productive and practical learning experience.
        </p>
    </div>

    <div class="highlight-grid">

        <div class="highlight-card">
            <div class="highlight-icon">🎓</div>
            <h3>Expert Sessions</h3>
            <p>Learn directly from professionals and experienced speakers.</p>
```

```
        </div>

        <div class="highlight-card">
            <div class="highlight-icon">👨‍💻</div>
            <h3>Hands-on Labs</h3>
            <p>Practice concepts through real-world technical activities.</p>
        </div>

        <div class="highlight-card">
            <div class="highlight-icon">📜</div>
            <h3>Certificate</h3>
            <p>Receive a certificate of participation after attending.</p>
        </div>

        <div class="highlight-card">
            <div class="highlight-icon">🤝</div>
            <h3>Networking</h3>
            <p>Connect with students, developers and industry professionals.</p>
        </div>

    </div>

</section>



<!-- Booking Section -->
<main id="booking" class="booking-section">

    <div class="section-heading">
        <span class="mini-label">REGISTRATION</span>
        <h2>Reserve Your Workshop Ticket</h2>
        <p>Fill in your details and customize your workshop experience.</p>
    </div>



    <form id="bookingForm" novalidate>

        <!-- Personal Details -->
        <div class="form-card">

            <div class="form-card-heading">
                <div class="step-number">01</div>
                <div>
                    <h3>Your Details</h3>
                    <p>Tell us who will be attending.</p>
                </div>
            </div>



            <div class="form-grid">

                <div class="form-group">
                    <label for="fullName">
                        Full Name <span>*</span>
                    </label>

                    <input
                        type="text"
                        id="fullName"
```

```
        placeholder="e.g. Sanskruti Gorle"
        autocomplete="name"
    >

    <small class="error-message" id="fullNameError"></small>
</div>



<div class="form-group">
    <label for="email">
        Email Address <span>*</span>
    </label>

    <input
        type="email"
        id="email"
        placeholder="e.g. student@gmail.com"
        autocomplete="email"
    >

    <small class="error-message" id="emailError"></small>
</div>



<div class="form-group">
    <label for="phone">
        Phone Number <span>*</span>
    </label>

    <input
        type="tel"
        id="phone"
        placeholder="10-digit mobile number"
        maxlength="10"
    >

    <small class="error-message" id="phoneError"></small>
</div>



<div class="form-group">
    <label for="studentId">
        Student ID / PRN <span>*</span>
    </label>

    <input
        type="text"
        id="studentId"
        placeholder="e.g. 24070521025"
    >

    <small class="error-message" id="studentIdError"></small>
</div>



<div class="form-group full-width">
    <label for="college">
        College / Organization <span>*</span>
    </label>
```

```
            <input
                type="text"
                id="college"
                placeholder="Enter your college or organization"
            >

            <small class="error-message" id="collegeError"></small>
        </div>

    </div>

</div>



<!-- Ticket Selection -->
<div class="form-card">

    <div class="form-card-heading">
        <div class="step-number">02</div>

        <div>
            <h3>Choose Your Ticket</h3>
            <p>Select the ticket that suits your workshop experience.</p>
        </div>
    </div>



    <div class="ticket-grid">

        <label class="ticket-option">

            <input
                type="radio"
                name="ticket"
                value="student"
                data-price="499"
                checked
            >

            <div class="ticket-content">
                <div class="ticket-top">
                    <span class="ticket-icon">🎓</span>
                    <span class="recommended">POPULAR</span>
                </div>

                <h3>Student Pass</h3>

                <p>
                    Full workshop access designed for students.
                </p>

                <strong>₹499</strong>
            </div>

        </label>



        <label class="ticket-option">
```

```
        <input
            type="radio"
            name="ticket"
            value="professional"
            data-price="999"
        >

        <div class="ticket-content">
            <div class="ticket-top">
                <span class="ticket-icon">💼</span>
            </div>

            <h3>Professional Pass</h3>

            <p>
                Workshop access with professional networking benefits.
            </p>

            <strong>₹999</strong>
        </div>

    </label>



    <label class="ticket-option">

        <input
            type="radio"
            name="ticket"
            value="premium"
            data-price="1499"
        >

        <div class="ticket-content">
            <div class="ticket-top">
                <span class="ticket-icon">⭐</span>
            </div>

            <h3>Premium Pass</h3>

            <p>
                Premium seating, kit and exclusive networking access.
            </p>

            <strong>₹1499</strong>
        </div>

    </label>

</div>



<div class="small-grid">

    <div class="form-group">
        <label for="quantity">
            Number of Tickets
        </label>
```

```
            <select id="quantity">

                <option value="1">1 Ticket</option>
                <option value="2">2 Tickets</option>
                <option value="3">3 Tickets</option>
                <option value="4">4 Tickets</option>
                <option value="5">5 Tickets</option>

            </select>
        </div>



        <div class="form-group">
            <label for="session">
                Preferred Session <span>*</span>
            </label>

            <select id="session">

                <option value="">
                    Select a session
                </option>

                <option value="ai">
                    AI & Machine Learning
                </option>

                <option value="web">
                    Modern Web Development
                </option>

                <option value="cloud">
                    Cloud & DevOps
                </option>

                <option value="cyber">
                    Cybersecurity Basics
                </option>

            </select>

            <small class="error-message" id="sessionError"></small>

        </div>

    </div>

</div>



<!-- Add-ons -->
<div class="form-card">

    <div class="form-card-heading">
        <div class="step-number">03</div>

        <div>
            <h3>Customize Your Experience</h3>
            <p>Add optional workshop benefits.</p>
```

```
    </div>
</div>


<div class="addon-list">

    <label class="addon">

        <input
            type="checkbox"
            class="addon-checkbox"
            value="Learning Kit"
            data-price="199"
        >

        <div class="addon-info">
            <span class="addon-icon">📦</span>

            <div>
                <strong>Learning Kit</strong>
                <small>Notebook, resources and workshop material</small>
            </div>
        </div>

        <strong class="addon-price">+₹199</strong>

    </label>


    <label class="addon">

        <input
            type="checkbox"
            class="addon-checkbox"
            value="Lunch"
            data-price="149"
        >

        <div class="addon-info">
            <span class="addon-icon">🍱</span>

            <div>
                <strong>Lunch</strong>
                <small>Vegetarian lunch and refreshments</small>
            </div>
        </div>

        <strong class="addon-price">+₹149</strong>

    </label>


    <label class="addon">

        <input
            type="checkbox"
            class="addon-checkbox"
            value="Certificate Upgrade"
            data-price="99"
```

```
            >

            <div class="addon-info">
                <span class="addon-icon">🏆</span>

                <div>
                    <strong>Certificate Upgrade</strong>
                    <small>Premium printed certificate</small>
                </div>
            </div>

            <strong class="addon-price">+₹99</strong>

        </label>

    </div>


    <div class="preference-row">

        <label for="foodPreference">
            🍽 Food Preference
        </label>

        <select id="foodPreference">

            <option value="Not Required">
                Not Required
            </option>

            <option value="Vegetarian">
                Vegetarian
            </option>

            <option value="Jain">
                Jain
            </option>

        </select>

    </div>

</div>


<!-- Additional Attendees -->
<div class="form-card">

    <div class="form-card-heading">
        <div class="step-number">04</div>

        <div>
            <h3>Additional Attendees</h3>
            <p>Add companions attending with you.</p>
        </div>
    </div>
```

```
    <div class="attendee-input">

        <input
            type="text"
            id="attendeeName"
            placeholder="Enter attendee name"
        >

        <button
            type="button"
            id="addAttendee"
            class="secondary-btn"
        >
            + Add Attendee
        </button>

    </div>

    <small class="helper-text">
        You can edit or remove an attendee before confirming your booking.
    </small>



    <div id="attendeeList" class="attendee-list">
        <!-- Dynamic attendees appear here -->
    </div>

</div>



<!-- Promo -->
<div class="form-card">

    <div class="form-card-heading">
        <div class="step-number">05</div>

        <div>
            <h3>Promo Code</h3>
            <p>Have a workshop discount code?</p>
        </div>
    </div>



    <div class="promo-row">

        <input
            type="text"
            id="promoCode"
            placeholder="Enter promo code"
            maxlength="20"
        >

        <button
            type="button"
            id="applyPromo"
            class="secondary-btn"
        >
            Apply Code
        </button>
```

```
    </div>

    <small id="promoMessage" class="promo-message"></small>

</div>



<!-- Booking Summary -->
<div class="summary-card" id="summary">

    <div class="summary-heading">

        <div>
            <span class="mini-label">ORDER SUMMARY</span>
            <h2>Your Workshop Booking</h2>
        </div>

        <span id="bookingStatus" class="status-badge">
            Draft
        </span>

    </div>



    <div id="selectedItems" class="selected-items">
        <!-- Generated dynamically -->
    </div>



    <div class="price-breakdown">

        <div>
            <span>Ticket Subtotal</span>
            <strong id="ticketSubtotal">₹499</strong>
        </div>

        <div>
            <span>Add-ons</span>
            <strong id="addonSubtotal">₹0</strong>
        </div>

        <div id="discountRow" class="discount-row hidden">
            <span>Discount</span>
            <strong id="discountAmount">-₹0</strong>
        </div>

        <div class="total-row">
            <span>Total</span>
            <strong id="grandTotal">₹499</strong>
        </div>

    </div>



    <div class="terms">

        <label>
```

```
                    <input
                        type="checkbox"
                        id="terms"
                    >

                    <span>
                        I confirm that the information provided is correct and
                        I agree to the workshop booking terms.
                    </span>

                </label>

                <small class="error-message" id="termsError"></small>

            </div>



            <div class="form-actions">

                <button
                    type="reset"
                    class="reset-btn"
                >
                    Reset
                </button>

                <button
                    type="submit"
                    class="primary-btn"
                >
                    Confirm Booking →
                </button>

            </div>

        </div>

    </form>

</main>



<!-- Confirmation Modal -->
<div
    id="confirmationModal"
    class="modal hidden"
>

    <div class="modal-content">

        <button
            id="closeModal"
            class="close-modal"
            type="button"
        >
            ×
        </button>

        <div class="success-icon">
```

```
            ✓
        </div>

        <span class="mini-label">BOOKING CONFIRMED</span>

        <h2>You're Registered! 🎉</h2>

        <p id="confirmationMessage">
            Your workshop ticket has been successfully booked.
        </p>



        <div class="registration-box">

            <small>REGISTRATION ID</small>

            <strong id="registrationId">
                TN-000000
            </strong>

        </div>



        <div class="confirmation-details" id="confirmationDetails">
            <!-- Dynamic confirmation details -->
        </div>



        <button
            id="doneButton"
            type="button"
            class="primary-btn full-btn"
        >
            Done
        </button>

    </div>

</div>



<!-- Footer -->
<footer>

    <div>
        <strong>TechNova</strong>
        <span>Technical Workshop 2026</span>
    </div>

    <p>
        Learn. Build. Innovate.
    </p>

    <small>
        © 2026 TechNova Workshop
    </small>

</footer>
```

```
    <!-- External JavaScript -->
    <script src="script.js"></script>

</body>
</html>
```

style.css

```
/* ================================
   TECHNOVA WORKSHOP
   PRACTICAL 10 - Q1
================================ */

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}



:root {
    --primary: #6c5ce7;
    --primary-dark: #5142c5;
    --secondary: #00b894;

    --bg: #f6f7fb;
    --surface: #ffffff;
    --surface-soft: #f1f2f8;

    --text: #202033;
    --muted: #73758a;

    --border: #e3e4ed;

    --danger: #e74c3c;
    --success: #159957;

    --shadow: 0 15px 45px rgba(38, 35, 80, 0.08);
}



body {
    font-family:
        Inter,
        "Segoe UI",
        Arial,
        sans-serif;

    background: var(--bg);
    color: var(--text);

    line-height: 1.6;

    transition:
        background 0.3s ease,
        color 0.3s ease;
}
```

```
html {
    scroll-behavior: smooth;
}



button,
input,
select {
    font: inherit;
}



/* ================================
   HEADER
================================ */

.header {
    position: sticky;
    top: 0;

    z-index: 100;

    display: flex;
    justify-content: space-between;
    align-items: center;

    padding: 16px 7%;

    background: rgba(255, 255, 255, 0.92);

    backdrop-filter: blur(15px);

    border-bottom: 1px solid var(--border);
}



.logo {
    display: flex;
    align-items: center;
    gap: 12px;
}



.logo-icon {
    width: 44px;
    height: 44px;

    display: grid;
    place-items: center;

    border-radius: 13px;

    background: linear-gradient(
        135deg,
        var(--primary),
        #9b8cff
    );
```

```
    color: white;

    font-size: 21px;

    box-shadow: 0 8px 20px rgba(108, 92, 231, 0.25);
}



.logo h2 {
    font-size: 18px;
    line-height: 1.1;
}



.logo small {
    color: var(--muted);
    font-size: 10px;
}



nav {
    display: flex;
    align-items: center;
    gap: 28px;
}



nav a {
    color: var(--text);

    text-decoration: none;

    font-size: 14px;
    font-weight: 600;

    transition: 0.2s;
}



nav a:hover {
    color: var(--primary);
}



.theme-btn {
    border: none;

    width: 40px;
    height: 40px;

    border-radius: 50%;

    background: var(--surface-soft);

    cursor: pointer;

    transition: 0.2s;
```

```
}


.theme-btn:hover {
    transform: rotate(15deg);
}


/* ================================
   HERO
================================ */

.hero {
    min-height: 600px;

    display: flex;
    align-items: center;
    justify-content: space-between;

    gap: 50px;

    padding: 80px 8%;

    overflow: hidden;

    background:
        radial-gradient(
            circle at 80% 30%,
            rgba(108, 92, 231, 0.13),
            transparent 30%
        );
}


.hero-content {
    max-width: 700px;
}


.badge,
.mini-label {
    display: inline-block;

    color: var(--primary);

    font-size: 11px;

    font-weight: 800;

    letter-spacing: 1.5px;
}


.badge {
    padding: 8px 14px;

    border-radius: 30px;
```

```
    background: rgba(108, 92, 231, 0.1);

    margin-bottom: 18px;
}



.hero h1 {
    font-size: clamp(42px, 6vw, 76px);

    line-height: 1.02;

    letter-spacing: -3px;

    margin-bottom: 25px;
}



.hero h1 span {
    display: block;

    color: var(--primary);
}



.hero-content > p {
    max-width: 600px;

    color: var(--muted);

    font-size: 17px;

    margin-bottom: 35px;
}



.hero-info {
    display: flex;
    flex-wrap: wrap;

    gap: 18px;

    margin-bottom: 35px;
}



.hero-info > div {
    display: flex;
    align-items: center;

    gap: 10px;

    padding: 13px 15px;

    border: 1px solid var(--border);

    border-radius: 14px;
```

```
    background: var(--surface);
}



.hero-info span {
    font-size: 20px;
}



.hero-info small,
.hero-info strong {
    display: block;
}



.hero-info small {
    color: var(--muted);

    font-size: 10px;
}



.hero-info strong {
    font-size: 12px;
}



.primary-btn {
    display: inline-flex;

    justify-content: center;
    align-items: center;

    gap: 8px;

    padding: 14px 24px;

    border: none;

    border-radius: 12px;

    background: linear-gradient(
        135deg,
        var(--primary),
        #8174ee
    );

    color: white;

    font-weight: 700;

    text-decoration: none;

    cursor: pointer;

    box-shadow:
        0 10px 25px rgba(108, 92, 231, 0.2);
```

```
    transition:
        transform 0.2s,
        box-shadow 0.2s;
}



.primary-btn:hover {
    transform: translateY(-2px);

    box-shadow:
        0 14px 30px rgba(108, 92, 231, 0.3);
}



.hero-visual {
    width: 400px;
    height: 400px;

    position: relative;

    display: grid;
    place-items: center;
}



.tech-circle {
    width: 270px;
    height: 270px;

    display: grid;
    place-items: center;

    border-radius: 50%;

    background:
        radial-gradient(
            circle,
            rgba(108, 92, 231, 0.2),
            rgba(108, 92, 231, 0.05)
        );

    border: 1px solid rgba(108, 92, 231, 0.2);
}



.circle-inner {
    width: 190px;
    height: 190px;

    display: flex;
    flex-direction: column;

    justify-content: center;
    align-items: center;

    border-radius: 50%;

    background: var(--surface);
```

```
    box-shadow: var(--shadow);
}



.circle-inner span {
    font-size: 48px;
}



.circle-inner strong {
    font-size: 22px;

    color: var(--primary);
}



.circle-inner small {
    color: var(--muted);

    letter-spacing: 3px;

    font-size: 9px;
}



.floating-card {
    position: absolute;

    display: flex;
    align-items: center;

    gap: 10px;

    padding: 13px 17px;

    border-radius: 15px;

    background: var(--surface);

    box-shadow: var(--shadow);

    border: 1px solid var(--border);
}



.floating-card > span {
    font-size: 25px;
}



.floating-card strong,
.floating-card small {
    display: block;
}
```

```
.floating-card strong {
    font-size: 12px;
}



.floating-card small {
    color: var(--muted);

    font-size: 10px;
}



.card-one {
    top: 35px;
    left: 0;
}



.card-two {
    right: 0;
    bottom: 40px;
}



/* ================================
   GENERAL SECTIONS
================================ */

.highlights,
.booking-section {
    padding: 85px 8%;
}



.section-heading {
    text-align: center;

    max-width: 700px;

    margin: 0 auto 45px;
}



.section-heading h2 {
    font-size: 36px;

    margin: 7px 0 10px;
}



.section-heading p {
    color: var(--muted);
}



.highlight-grid {
    display: grid;
```

```
    grid-template-columns:
        repeat(4, 1fr);

    gap: 18px;
}


.highlight-card {
    padding: 28px;

    background: var(--surface);

    border: 1px solid var(--border);

    border-radius: 20px;

    transition: 0.25s;
}


.highlight-card:hover {
    transform: translateY(-5px);

    box-shadow: var(--shadow);
}


.highlight-icon {
    width: 50px;
    height: 50px;

    display: grid;
    place-items: center;

    border-radius: 14px;

    background: var(--surface-soft);

    font-size: 24px;

    margin-bottom: 18px;
}


.highlight-card h3 {
    margin-bottom: 7px;
}


.highlight-card p {
    color: var(--muted);

    font-size: 13px;
}


/* ================================
```

```
   FORM
================================ */

.booking-section {
    background:
        linear-gradient(
            180deg,
            var(--bg),
            #eef0f8
        );
}



.form-card,
.summary-card {
    max-width: 1050px;

    margin: 0 auto 22px;

    background: var(--surface);

    border: 1px solid var(--border);

    border-radius: 22px;

    padding: 30px;

    box-shadow: 0 8px 30px rgba(40, 40, 80, 0.04);
}



.form-card-heading {
    display: flex;

    align-items: center;

    gap: 14px;

    margin-bottom: 25px;
}



.step-number {
    width: 45px;
    height: 45px;

    flex-shrink: 0;

    display: grid;
    place-items: center;

    border-radius: 13px;

    background: rgba(108, 92, 231, 0.1);

    color: var(--primary);

    font-size: 12px;
```

```
    font-weight: 800;
}



.form-card-heading h3 {
    font-size: 19px;
}



.form-card-heading p {
    color: var(--muted);

    font-size: 12px;
}



.form-grid {
    display: grid;

    grid-template-columns:
        repeat(2, 1fr);

    gap: 20px;
}



.full-width {
    grid-column: 1 / -1;
}



.form-group label {
    display: block;

    margin-bottom: 7px;

    font-size: 13px;

    font-weight: 700;
}



.form-group label span {
    color: var(--danger);
}



input,
select {
    width: 100%;

    border: 1px solid var(--border);

    background: var(--surface);

    color: var(--text);
```

```
    padding: 13px 14px;

    border-radius: 11px;

    outline: none;

    transition:
        border 0.2s,
        box-shadow 0.2s,
        background 0.2s;
}



input::placeholder {
    color: #aaaabd;
}



input:focus,
select:focus {
    border-color: var(--primary);

    box-shadow:
        0 0 0 4px rgba(108, 92, 231, 0.08);
}



.input-error {
    border-color: var(--danger) !important;

    box-shadow:
        0 0 0 4px rgba(231, 76, 60, 0.08) !important;
}



.input-success {
    border-color: var(--success) !important;
}



.error-message {
    display: block;

    min-height: 17px;

    margin-top: 4px;

    color: var(--danger);

    font-size: 11px;
}



/* ================================
   TICKETS
================================ */
```

```
.ticket-grid {
    display: grid;

    grid-template-columns:
        repeat(3, 1fr);

    gap: 15px;

    margin-bottom: 22px;
}



.ticket-option {
    cursor: pointer;
}



.ticket-option input {
    position: absolute;

    opacity: 0;

    pointer-events: none;
}



.ticket-content {
    height: 100%;

    padding: 22px;

    border: 2px solid var(--border);

    border-radius: 17px;

    transition: 0.2s;
}



.ticket-option input:checked + .ticket-content {
    border-color: var(--primary);

    background:
        rgba(108, 92, 231, 0.05);

    box-shadow:
        0 8px 25px rgba(108, 92, 231, 0.08);
}



.ticket-top {
    display: flex;

    justify-content: space-between;

    margin-bottom: 18px;
}
```

```
.ticket-icon {
    font-size: 27px;
}



.recommended {
    padding: 5px 8px;

    background: rgba(0, 184, 148, 0.1);

    color: var(--secondary);

    border-radius: 7px;

    font-size: 8px;

    font-weight: 800;
}



.ticket-content h3 {
    margin-bottom: 7px;
}



.ticket-content p {
    min-height: 48px;

    color: var(--muted);

    font-size: 12px;

    margin-bottom: 12px;
}



.ticket-content > strong {
    color: var(--primary);

    font-size: 21px;
}



.small-grid {
    display: grid;

    grid-template-columns:
        repeat(2, 1fr);

    gap: 20px;
}



/* ================================
   ADDONS
```

```
================================ */

.addon-list {
    display: grid;

    gap: 10px;
}



.addon {
    display: flex;

    align-items: center;

    gap: 15px;

    padding: 15px;

    border: 1px solid var(--border);

    border-radius: 14px;

    cursor: pointer;

    transition: 0.2s;
}



.addon:hover {
    border-color: var(--primary);
}



.addon input {
    width: auto;
}



.addon-info {
    display: flex;

    align-items: center;

    gap: 12px;

    flex: 1;
}



.addon-icon {
    width: 38px;
    height: 38px;

    display: grid;
    place-items: center;

    border-radius: 10px;
```

```
    background: var(--surface-soft);
}



.addon-info strong,
.addon-info small {
    display: block;
}



.addon-info small {
    color: var(--muted);

    font-size: 11px;
}



.addon-price {
    color: var(--primary);

    font-size: 13px;
}



.preference-row {
    display: flex;

    align-items: center;

    justify-content: space-between;

    gap: 20px;

    margin-top: 20px;

    padding-top: 20px;

    border-top: 1px solid var(--border);
}



.preference-row select {
    max-width: 250px;
}



/* ================================
   ATTENDEES
================================ */

.attendee-input {
    display: flex;

    gap: 10px;
}
```

```
.secondary-btn {
    border: 1px solid var(--primary);

    background: rgba(108, 92, 231, 0.07);

    color: var(--primary);

    padding: 12px 18px;

    border-radius: 10px;

    font-weight: 700;

    cursor: pointer;

    white-space: nowrap;

    transition: 0.2s;
}


.secondary-btn:hover {
    background: var(--primary);

    color: white;
}


.helper-text {
    display: block;

    margin-top: 8px;

    color: var(--muted);

    font-size: 11px;
}


.attendee-list {
    display: grid;

    gap: 9px;

    margin-top: 18px;
}


.attendee-item {
    display: flex;

    align-items: center;

    justify-content: space-between;

    gap: 10px;
```

```
    padding: 12px 14px;

    border-radius: 11px;

    background: var(--surface-soft);
}



.attendee-name {
    font-size: 13px;

    font-weight: 600;
}



.attendee-actions {
    display: flex;

    gap: 7px;
}



.attendee-actions button {
    border: none;

    padding: 6px 10px;

    border-radius: 7px;

    cursor: pointer;

    font-size: 11px;
}



.edit-btn {
    background: rgba(108, 92, 231, 0.12);

    color: var(--primary);
}



.delete-btn {
    background: rgba(231, 76, 60, 0.1);

    color: var(--danger);
}



/* ================================
   PROMO
================================ */

.promo-row {
    display: flex;
```

```
    gap: 10px;
}


.promo-message {
    display: block;

    margin-top: 8px;

    font-size: 12px;
}


.promo-success {
    color: var(--success);
}


.promo-error {
    color: var(--danger);
}


/* ================================
   SUMMARY
================================ */

.summary-card {
    margin-top: 30px;

    border: 2px solid rgba(108, 92, 231, 0.15);
}


.summary-heading {
    display: flex;

    justify-content: space-between;

    align-items: center;

    margin-bottom: 25px;
}


.summary-heading h2 {
    margin-top: 5px;
}


.status-badge {
    padding: 7px 12px;

    border-radius: 30px;

    background: rgba(255, 193, 7, 0.12);
```

```
    color: #b07a00;

    font-size: 10px;

    font-weight: 800;
}



.selected-items {
    display: grid;

    gap: 10px;

    margin-bottom: 20px;
}



.selected-item {
    display: flex;

    justify-content: space-between;

    align-items: center;

    padding: 13px 15px;

    background: var(--surface-soft);

    border-radius: 11px;

    font-size: 13px;
}



.selected-item span {
    color: var(--muted);
}



.price-breakdown {
    border-top: 1px solid var(--border);
}



.price-breakdown > div {
    display: flex;

    justify-content: space-between;

    padding: 11px 0;

    font-size: 13px;
}



.discount-row {
    color: var(--success);
```

```
}


.hidden {
    display: none !important;
}


.total-row {
    margin-top: 8px;

    padding-top: 17px !important;

    border-top: 1px solid var(--border);

    font-size: 20px !important;
}


.total-row strong {
    color: var(--primary);

    font-size: 24px;
}


.terms {
    margin-top: 20px;

    padding: 15px;

    background: var(--surface-soft);

    border-radius: 12px;
}


.terms label {
    display: flex;

    align-items: flex-start;

    gap: 10px;

    font-size: 12px;

    cursor: pointer;
}


.terms input {
    width: auto;

    margin-top: 4px;
}
```

```
.form-actions {
    display: flex;

    justify-content: flex-end;

    gap: 12px;

    margin-top: 25px;
}



.reset-btn {
    padding: 13px 22px;

    border: 1px solid var(--border);

    background: transparent;

    color: var(--muted);

    border-radius: 11px;

    cursor: pointer;

    font-weight: 700;
}



.reset-btn:hover {
    background: var(--surface-soft);
}



/* ================================
   MODAL
================================ */

.modal {
    position: fixed;

    inset: 0;

    z-index: 500;

    display: grid;

    place-items: center;

    padding: 20px;

    background:
        rgba(15, 15, 30, 0.65);

    backdrop-filter: blur(8px);
}
```

```
.modal.hidden {
    display: none;
}



.modal-content {
    position: relative;

    width: min(500px, 100%);

    padding: 40px;

    text-align: center;

    background: var(--surface);

    border-radius: 25px;

    box-shadow:
        0 30px 80px rgba(0, 0, 0, 0.25);
}



.close-modal {
    position: absolute;

    right: 18px;
    top: 15px;

    width: 35px;
    height: 35px;

    border: none;

    border-radius: 50%;

    background: var(--surface-soft);

    color: var(--muted);

    font-size: 24px;

    cursor: pointer;
}



.success-icon {
    width: 70px;
    height: 70px;

    display: grid;
    place-items: center;

    margin: 0 auto 20px;

    border-radius: 50%;
```

```
    background: rgba(21, 153, 87, 0.12);

    color: var(--success);

    font-size: 35px;

    font-weight: bold;
}


.modal-content h2 {
    margin: 8px 0;

    font-size: 28px;
}


.modal-content > p {
    color: var(--muted);

    font-size: 13px;

    margin-bottom: 22px;
}


.registration-box {
    padding: 16px;

    background: var(--surface-soft);

    border-radius: 13px;

    margin-bottom: 15px;
}


.registration-box small,
.registration-box strong {
    display: block;
}


.registration-box small {
    color: var(--muted);

    font-size: 9px;

    letter-spacing: 2px;
}


.registration-box strong {
    margin-top: 5px;

    color: var(--primary);
```

```
    font-size: 22px;

    letter-spacing: 2px;
}


.confirmation-details {
    display: grid;

    gap: 5px;

    margin-bottom: 20px;

    color: var(--muted);

    font-size: 12px;
}


.full-btn {
    width: 100%;
}


/* ================================
   FOOTER
================================ */

footer {
    display: flex;

    justify-content: space-between;

    align-items: center;

    gap: 20px;

    padding: 25px 8%;

    background: #171625;

    color: white;
}


footer strong,
footer span {
    display: block;
}


footer span,
footer small {
    color: #9694a8;

    font-size: 11px;
}
```

```
footer p {
    color: #c9c7d8;

    font-size: 12px;
}



/* ================================
   DARK MODE
================================ */

body.dark-mode {
    --bg: #11111a;

    --surface: #1a1a27;

    --surface-soft: #242433;

    --text: #f2f2fa;

    --muted: #a2a1b4;

    --border: #303043;
}



body.dark-mode .header {
    background: rgba(17, 17, 26, 0.9);
}



body.dark-mode .booking-section {
    background: #11111a;
}



body.dark-mode .ticket-content {
    background: #1a1a27;
}



body.dark-mode .hero {
    background:
        radial-gradient(
            circle at 80% 30%,
            rgba(108, 92, 231, 0.2),
            transparent 30%
        );
}



/* ================================
   RESPONSIVE
================================ */

@media (max-width: 950px) {
```

```
    .hero {
        padding-top: 55px;

        flex-direction: column;

        text-align: center;
    }


    .hero-content {
        max-width: 750px;
    }


    .hero-content > p {
        margin-left: auto;
        margin-right: auto;
    }


    .hero-info {
        justify-content: center;
    }


    .hero-visual {
        width: 350px;
    }


    .highlight-grid {
        grid-template-columns:
            repeat(2, 1fr);
    }


    .ticket-grid {
        grid-template-columns:
            1fr;
    }

}


@media (max-width: 700px) {

    .header {
        padding: 13px 5%;
    }


    nav a {
        display: none;
    }


    .hero,
```

```
.highlights,
.booking-section {
    padding-left: 5%;
    padding-right: 5%;
}



.hero h1 {
    font-size: 45px;

    letter-spacing: -2px;
}



.hero-info {
    flex-direction: column;
}



.hero-info > div {
    justify-content: flex-start;
}



.highlight-grid,
.form-grid,
.small-grid {
    grid-template-columns: 1fr;
}



.full-width {
    grid-column: auto;
}



.form-card,
.summary-card {
    padding: 20px;

    border-radius: 17px;
}



.attendee-input,
.promo-row {
    flex-direction: column;
}



.preference-row {
    flex-direction: column;

    align-items: stretch;
}



.preference-row select {
```

```
        max-width: none;
    }


    .form-actions {
        flex-direction: column-reverse;
    }


    .form-actions button {
        width: 100%;
    }


    footer {
        flex-direction: column;

        text-align: center;
    }

}
```

script.js

```
/* ======================================================
   TECHNOVA TECHNICAL WORKSHOP BOOKING
   PRACTICAL 10 - Q1

   Concepts integrated from Practical 1 to 9:
   P1 -> console, external JS, user information
   P2 -> var, let, const, template literals, destructuring
   P3 -> conditions and validation
   P4 -> functions, scope, closure, try-catch
   P5 -> arrays, objects, map, filter, find, reduce, includes
   P6 -> strings and regular expressions
   P7 -> DOM traversal and dynamic DOM manipulation
   P8 -> events
   P9 -> localStorage and sessionStorage
====================================================== */



/* ======================================================
   P1 - CONSOLE METHODS
====================================================== */

console.log("================================");
console.log("TechNova Workshop Booking Started");
console.log("================================");

console.info("JavaScript file loaded successfully.");



// Small user information object
const applicationInfo = {
    application: "Technical Workshop Ticket Booking",
    year: 2026,
    version: "1.0"
};

console.table(applicationInfo);
```

```
/* ====================================================
   P2 - VARIABLES
==================================================== */

// const -> values that should not be reassigned
const workshopName = "TechNova Technical Workshop";
const workshopDate = "18 October 2026";



// let -> values that change during application
let discount = 0;
let appliedPromo = "";



// var -> older JavaScript variable declaration
// Used here intentionally as part of Practical 2.
var bookingCount = 0;



/* ====================================================
   P5 - ARRAY + OBJECT DATA
==================================================== */

const ticketTypes = [
    {
        id: "student",
        name: "Student Pass",
        price: 499
    },

    {
        id: "professional",
        name: "Professional Pass",
        price: 999
    },

    {
        id: "premium",
        name: "Premium Pass",
        price: 1499
    }
];



const promoCodes = [
    {
        code: "TECH10",
        percentage: 10
    },

    {
        code: "STUDENT15",
        percentage: 15
    },

    {
        code: "WELCOME20",
        percentage: 20
```

```
    }
];


let attendees = [];


/* ======================================================
   P4 - CLOSURE
   Generates unique registration IDs.
====================================================== */

function createRegistrationIdGenerator() {

    let counter =
        Number(localStorage.getItem("bookingCounter")) || 0;

    return function generateRegistrationId() {

        counter++;

        localStorage.setItem(
            "bookingCounter",
            counter
        );

        return `TN-${String(counter).padStart(6, "0")}`;
    };
}


const generateRegistrationId =
    createRegistrationIdGenerator();


/* ======================================================
   DOM REFERENCES
====================================================== */

const bookingForm =
    document.getElementById("bookingForm");

const fullName =
    document.getElementById("fullName");

const email =
    document.getElementById("email");

const phone =
    document.getElementById("phone");

const studentId =
    document.getElementById("studentId");

const college =
    document.getElementById("college");

const session =
```

```
    document.getElementById("session");

const quantity =
    document.getElementById("quantity");

const foodPreference =
    document.getElementById("foodPreference");

const attendeeName =
    document.getElementById("attendeeName");

const attendeeList =
    document.getElementById("attendeeList");

const addAttendee =
    document.getElementById("addAttendee");

const promoCode =
    document.getElementById("promoCode");

const applyPromo =
    document.getElementById("applyPromo");

const promoMessage =
    document.getElementById("promoMessage");

const terms =
    document.getElementById("terms");

const themeToggle =
    document.getElementById("themeToggle");

const confirmationModal =
    document.getElementById("confirmationModal");

const closeModal =
    document.getElementById("closeModal");

const doneButton =
    document.getElementById("doneButton");



/* ====================================================
   P9 - LOAD THEME FROM LOCAL STORAGE
==================================================== */

function loadSavedTheme() {

    const savedTheme =
        localStorage.getItem("theme");

    if (savedTheme === "dark") {

        document.body.classList.add("dark-mode");

        themeToggle.textContent = "◯";

    } else {
```

```
        themeToggle.textContent = "🌜";
    }
}


loadSavedTheme();


/* ====================================================
   P9 - THEME TOGGLE
==================================================== */

themeToggle.addEventListener(
    "click",
    function () {

        document.body.classList.toggle("dark-mode");

        const isDark =
            document.body.classList.contains("dark-mode");

        localStorage.setItem(
            "theme",
            isDark ? "dark" : "light"
        );

        themeToggle.textContent =
            isDark ? "🌞" : "🌜";

        console.log(
            `Theme changed to ${isDark ? "dark" : "light"} mode.`
        );
    }
);


/* ====================================================
   P3 + P6
   REGULAR EXPRESSIONS
==================================================== */

const nameRegex =
    /^[A-Za-z]+(?:[ '-][A-Za-z]+)*$/;

const emailRegex =
    /^[^\s@]+@[^\s@]+\.[^\s@]{2,}$/;

const phoneRegex =
    /^[6-9]\d{9}$/;

const studentIdRegex =
    /^[A-Za-z0-9-]{5,20}$/;


/* ====================================================
   P4 - FUNCTION DECLARATION
==================================================== */
```

```
function showError(input, errorElement, message) {

    input.classList.remove("input-success");

    input.classList.add("input-error");

    errorElement.textContent = message;

    return false;
}


/* ======================================================
   FUNCTION EXPRESSION
====================================================== */

const showSuccess =
    function (input, errorElement) {

        input.classList.remove("input-error");

        input.classList.add("input-success");

        errorElement.textContent = "";

        return true;
    };


/* ======================================================
   GET ERROR ELEMENT
====================================================== */

function getErrorElement(inputId) {

    return document.getElementById(
        `${inputId}Error`
    );
}


/* ======================================================
   P3 + P6
   VALIDATION FUNCTIONS
====================================================== */

function validateName() {

    const value =
        fullName.value.trim();

    const error =
        getErrorElement("fullName");

    if (value === "") {

        return showError(
```

```
            fullName,
            error,
            "Please enter your full name."
        );
    }

    if (!nameRegex.test(value)) {

        return showError(
            fullName,
            error,
            "Please enter a valid name."
        );
    }

    return showSuccess(
        fullName,
        error
    );
}


function validateEmail() {

    const value =
        email.value.trim();

    const error =
        getErrorElement("email");

    if (value === "") {

        return showError(
            email,
            error,
            "Email address is required."
        );
    }

    if (!emailRegex.test(value)) {

        return showError(
            email,
            error,
            "Enter a valid email address."
        );
    }

    return showSuccess(
        email,
        error
    );
}


function validatePhone() {

    const value =
        phone.value.trim();
```

```
    const error =
        getErrorElement("phone");

    if (value === "") {

        return showError(
            phone,
            error,
            "Phone number is required."
        );
    }

    if (!phoneRegex.test(value)) {

        return showError(
            phone,
            error,
            "Enter a valid 10-digit mobile number."
        );
    }

    return showSuccess(
        phone,
        error
    );
}


function validateStudentId() {

    const value =
        studentId.value.trim();

    const error =
        getErrorElement("studentId");

    if (value === "") {

        return showError(
            studentId,
            error,
            "Student ID is required."
        );
    }

    if (!studentIdRegex.test(value)) {

        return showError(
            studentId,
            error,
            "Enter a valid student ID."
        );
    }

    return showSuccess(
        studentId,
        error
    );
}
```

```
function validateCollege() {

    const value =
        college.value.trim();

    const error =
        getErrorElement("college");

    if (value === "") {

        return showError(
            college,
            error,
            "College or organization is required."
        );
    }

    if (value.length < 3) {

        return showError(
            college,
            error,
            "Please enter a valid college name."
        );
    }

    return showSuccess(
        college,
        error
    );
}


function validateSession() {

    const error =
        document.getElementById("sessionError");

    if (session.value === "") {

        return showError(
            session,
            error,
            "Please select a preferred session."
        );
    }

    return showSuccess(
        session,
        error
    );
}


function validateTerms() {

    const error =
        document.getElementById("termsError");
```

```
    if (!terms.checked) {

        error.textContent =
            "Please accept the booking terms.";

        return false;
    }

    error.textContent = "";

    return true;
}



/* ======================================================
   P8 - INPUT EVENTS
====================================================== */

fullName.addEventListener(
    "input",
    function () {

        // P6 - string function
        fullName.value =
            fullName.value.replace(
                /[^A-Za-z '-]/g,
                ""
            );

        validateName();

        updateSummary();
    }
);



email.addEventListener(
    "input",
    function () {

        validateEmail();

        updateSummary();
    }
);



phone.addEventListener(
    "input",
    function () {

        // String method
        phone.value =
            phone.value.replace(
                /\D/g,
                ""
            );
```

```
        validatePhone();
    }
);


studentId.addEventListener(
    "input",
    function () {

        studentId.value =
            studentId.value.trim();

        validateStudentId();
    }
);


college.addEventListener(
    "input",
    validateCollege
);


/* ====================================================
   P8 - FOCUS / BLUR EVENTS
==================================================== */

const requiredInputs =
    [fullName, email, phone, studentId, college];


requiredInputs.forEach(
    function (input) {

        input.addEventListener(
            "focus",
            function () {

                input.parentElement
                    .classList.add("focused");

                console.log(
                    `${input.id} field focused.`
                );
            }
        );


        input.addEventListener(
            "blur",
            function () {

                input.parentElement
                    .classList.remove("focused");

                // Validate when user leaves field
                switch (input.id) {

                    case "fullName":
```

```
                        validateName();
                        break;

                    case "email":
                        validateEmail();
                        break;

                    case "phone":
                        validatePhone();
                        break;

                    case "studentId":
                        validateStudentId();
                        break;

                    case "college":
                        validateCollege();
                        break;
                }
            }
        );
    }
);


/* ======================================================
   P5 - FIND
   Find selected ticket object.
====================================================== */

function getSelectedTicket() {

    const selectedRadio =
        document.querySelector(
            'input[name="ticket"]:checked'
        );

    const ticketId =
        selectedRadio.value;

    return ticketTypes.find(
        ticket => ticket.id === ticketId
    );
}


/* ======================================================
   P2 - DESTRUCTURING
====================================================== */

function getTicketInformation() {

    const ticket =
        getSelectedTicket();

    const {
        name: ticketName,
        price: ticketPrice
    } = ticket;
```

```
    return {
        ticketName,
        ticketPrice
    };
}


/* ================================================
   P8 - CHANGE EVENTS
================================================ */

document
    .querySelectorAll('input[name="ticket"]')
    .forEach(
        radio => {

            radio.addEventListener(
                "change",
                function () {

                    const {
                        ticketName,
                        ticketPrice
                    } = getTicketInformation();

                    console.log(
                        `Ticket changed: ${ticketName} - ₹${ticketPrice}`
                    );

                    updateSummary();
                }
            );
        }
    );


quantity.addEventListener(
    "change",
    function () {

        sessionStorage.setItem(
            "ticketQuantity",
            quantity.value
        );

        updateSummary();

        console.log(
            `Ticket quantity changed to ${quantity.value}`
        );
    }
);


session.addEventListener(
    "change",
    function () {

        sessionStorage.setItem(
            "preferredSession",
```

```
            session.value
        );

        validateSession();

        updateSummary();
    }
);


foodPreference.addEventListener(
    "change",
    function () {

        sessionStorage.setItem(
            "foodPreference",
            foodPreference.value
        );

        updateSummary();
    }
);


/* ======================================================
   P5 - FILTER
   GET SELECTED ADD-ONS
====================================================== */

function getSelectedAddons() {

    const addonCheckboxes =
        document.querySelectorAll(
            ".addon-checkbox"
        );

    const addonArray =
        Array.from(addonCheckboxes);

    return addonArray
        .filter(
            checkbox => checkbox.checked
        )
        .map(
            checkbox => {

                return {
                    name: checkbox.value,
                    price: Number(
                        checkbox.dataset.price
                    )
                };
            }
        );
}


/* ======================================================
   P5 - REDUCE
   CALCULATE ADDON TOTAL
```

```
====================================================== */

function calculateAddonTotal(addons) {

    return addons.reduce(
        (total, addon) => {

            return total + addon.price;

        },
        0
    );
}



/* ======================================================
   P5 - MAP
   DISPLAY ADDONS
====================================================== */

document
    .querySelectorAll(".addon-checkbox")
    .forEach(
        checkbox => {

            checkbox.addEventListener(
                "change",
                function () {

                    const addons =
                        getSelectedAddons();

                    const names =
                        addons.map(
                            addon => addon.name
                        );

                    console.log(
                        "Selected add-ons:",
                        names
                    );

                    updateSummary();
                }
            );
        }
    );



/* ======================================================
   P6 - STRING FUNCTIONS
====================================================== */

function extractEmailDomain(emailAddress) {

    const parts =
        emailAddress
            .trim()
            .toLowerCase()
            .split("@");
```

```
    return parts[1] || "";
}


/* ======================================================
   P6 - includes()
====================================================== */

function isStudentEmail(emailAddress) {

    const domain =
        extractEmailDomain(emailAddress);

    return (
        domain.includes("edu") ||
        domain.includes("ac.")
    );
}


/* ======================================================
   P9 - SESSION STORAGE
   Save current user preferences.
====================================================== */

function saveSessionData() {

    const sessionData = {

        name: fullName.value,

        email: email.value,

        ticket:
            getSelectedTicket().id,

        quantity:
            quantity.value,

        workshopSession:
            session.value,

        food:
            foodPreference.value

    };


    sessionStorage.setItem(
        "workshopBooking",
        JSON.stringify(sessionData)
    );
}


/* ======================================================
   P9 - LOAD SESSION DATA
```

```
================================================== */

function loadSessionData() {

    try {

        const savedData =
            sessionStorage.getItem(
                "workshopBooking"
            );

        if (!savedData) {
            return;
        }

        const data =
            JSON.parse(savedData);

        // Restore only useful fields
        if (data.name) {
            fullName.value = data.name;
        }

        if (data.email) {
            email.value = data.email;
        }

        if (data.quantity) {
            quantity.value = data.quantity;
        }

        if (data.workshopSession) {
            session.value =
                data.workshopSession;
        }

        if (data.food) {
            foodPreference.value =
                data.food;
        }



        if (data.ticket) {

            const radio =
                document.querySelector(
                    `input[name="ticket"][value="${data.ticket}"]`
                );

            if (radio) {
                radio.checked = true;
            }
        }



        console.info(
            "Previous session preferences restored."
        );
```

```
    } catch (error) {

        console.error(
            "Could not restore session data:",
            error
        );

    }
}


loadSessionData();


/* ======================================================
   UPDATE SUMMARY
====================================================== */

function updateSummary() {

    const {
        ticketName,
        ticketPrice
    } = getTicketInformation();


    const ticketQuantity =
        Number(quantity.value);


    const ticketSubtotal =
        ticketPrice * ticketQuantity;


    const addons =
        getSelectedAddons();


    const addonSubtotal =
        calculateAddonTotal(addons);


    const subtotal =
        ticketSubtotal + addonSubtotal;


    let discountAmount = 0;


    if (discount > 0) {

        discountAmount =
            Math.round(
                subtotal * discount / 100
            );
    }
```

```
const grandTotal =
    subtotal - discountAmount;



document.getElementById(
    "ticketSubtotal"
).textContent =
    `₹${ticketSubtotal}`;



document.getElementById(
    "addonSubtotal"
).textContent =
    `₹${addonSubtotal}`;



document.getElementById(
    "discountAmount"
).textContent =
    `-₹${discountAmount}`;



document.getElementById(
    "grandTotal"
).textContent =
    `₹${grandTotal}`;



/* ================================
   Dynamic selected item DOM
================================ */

const selectedItems =
    document.getElementById(
        "selectedItems"
    );



selectedItems.innerHTML = "";



const ticketElement =
    document.createElement("div");

ticketElement.className =
    "selected-item";



ticketElement.innerHTML = `
    <span>
        ${ticketName} × ${ticketQuantity}
    </span>

    <strong>
        ₹${ticketSubtotal}
    </strong>
`;
```

```
    selectedItems.appendChild(
        ticketElement
    );



    // Add-ons are dynamically inserted
    addons.forEach(
        addon => {

            const addonElement =
                document.createElement(
                    "div"
                );

            addonElement.className =
                "selected-item";



            addonElement.innerHTML = `
                <span>
                    ${addon.name}
                </span>

                <strong>
                    ₹${addon.price}
                </strong>
            `;



            selectedItems.appendChild(
                addonElement
            );
        }
    );



    // Save current state
    saveSessionData();
}



/* ====================================================
   PROMO CODE
==================================================== */

applyPromo.addEventListener(
    "click",
    function () {

        const enteredCode =
            promoCode.value
                .trim()
                .toUpperCase();



        if (enteredCode === "") {
```

```
    promoMessage.textContent =
        "Please enter a promo code.";

    promoMessage.className =
        "promo-message promo-error";

    return;
}



/*
    P5 - FIND
    Search promo object in array.
*/

const promo =
    promoCodes.find(
        item =>
            item.code === enteredCode
    );



if (promo) {

    discount =
        promo.percentage;

    appliedPromo =
        enteredCode;



    promoMessage.textContent =
        `${promo.percentage}% discount applied successfully!`;



    promoMessage.className =
        "promo-message promo-success";



    document
        .getElementById("discountRow")
        .classList.remove("hidden");



    console.log(
        `Promo code ${enteredCode} applied.`
    );

} else {

    discount = 0;

    appliedPromo = "";



    promoMessage.textContent =
        "Invalid promo code.";
```

```
            promoMessage.className =
                "promo-message promo-error";


            document
                .getElementById("discountRow")
                .classList.add("hidden");


            console.warn(
                "Invalid promo code entered."
            );
        }


        updateSummary();
    }
);


/* ================================================
   P7 - ADD ATTENDEE
================================================ */

addAttendee.addEventListener(
    "click",
    function () {

        const name =
            attendeeName.value.trim();


        if (name === "") {

            attendeeName.classList.add(
                "input-error"
            );

            attendeeName.focus();

            return;
        }


        if (!nameRegex.test(name)) {

            attendeeName.classList.add(
                "input-error"
            );

            return;
        }


        /*
            P5 - includes()
```

```
            Prevent duplicate attendee names.
        */

        const alreadyExists =
            attendees
                .map(
                    attendee =>
                        attendee.name.toLowerCase()
                )
                .includes(
                    name.toLowerCase()
                );


        if (alreadyExists) {

            alert(
                "This attendee has already been added."
            );

            return;
        }



        attendeeName.classList.remove(
            "input-error"
        );



        /*
            P5 - OBJECT
        */

        attendees.push({
            id: Date.now(),
            name: name
        });



        attendeeName.value = "";



        renderAttendees();



        console.log(
            "Attendee added:",
            name
        );
    }
);



/* ====================================================
   P7 - CREATE + APPEND DOM
==================================================== */

function renderAttendees() {
```

```
    attendeeList.innerHTML = "";


    attendees.forEach(
        attendee => {

            const item =
                document.createElement(
                    "div"
                );

            item.className =
                "attendee-item";



            item.dataset.id =
                attendee.id;



            item.innerHTML = `

                <span class="attendee-name">
                    ${attendee.name}
                </span>

                <div class="attendee-actions">

                    <button
                        type="button"
                        class="edit-btn"
                        data-action="edit"
                        data-id="${attendee.id}"
                    >
                        Edit
                    </button>

                    <button
                        type="button"
                        class="delete-btn"
                        data-action="delete"
                        data-id="${attendee.id}"
                    >
                        Delete
                    </button>

                </div>
            `;



            attendeeList.appendChild(
                item
            );
        }
    );
}


/* ================================================
```

```
   P7 - EVENT DELEGATION
   EDIT + DELETE
================================================== */

attendeeList.addEventListener(
    "click",
    function (event) {

        const button =
            event.target.closest("button");



        if (!button) {
            return;
        }



        const id =
            Number(button.dataset.id);



        const action =
            button.dataset.action;



        if (action === "delete") {

            /*
                P5 - FILTER
                Remove selected attendee.
            */

            attendees =
                attendees.filter(
                    attendee =>
                        attendee.id !== id
                );



            renderAttendees();



            console.log(
                `Attendee ${id} deleted.`
            );
        }



        if (action === "edit") {

            const attendee =
                attendees.find(
                    item =>
                        item.id === id
                );



            if (!attendee) {
```

```
                return;
            }


            const updatedName =
                prompt(
                    "Edit attendee name:",
                    attendee.name
                );


            if (
                updatedName !== null &&
                updatedName.trim() !== ""
            ) {

                if (
                    nameRegex.test(
                        updatedName.trim()
                    )
                ) {

                    attendee.name =
                        updatedName.trim();


                    renderAttendees();


                    console.log(
                        "Attendee updated:",
                        attendee.name
                    );

                } else {

                    alert(
                        "Please enter a valid name."
                    );
                }
            }
        }
    }
);


/* ==================================================
   P8 - ATTENDEE INPUT EVENT
================================================== */

attendeeName.addEventListener(
    "input",
    function () {

        if (nameRegex.test(
            attendeeName.value.trim()
        )) {

            attendeeName.classList.remove(
```

```
                "input-error"
            );
        }
    }
);


/* ================================================
   P8 - FORM SUBMIT
================================================ */

bookingForm.addEventListener(
    "submit",
    function (event) {

        event.preventDefault();



        console.log(
            "Booking form submitted."
        );



        /*
            P3 - Validation
        */

        const validName =
            validateName();

        const validEmail =
            validateEmail();

        const validPhone =
            validatePhone();

        const validStudentId =
            validateStudentId();

        const validCollege =
            validateCollege();

        const validSession =
            validateSession();

        const validTerms =
            validateTerms();



        const formIsValid =
            validName &&
            validEmail &&
            validPhone &&
            validStudentId &&
            validCollege &&
            validSession &&
            validTerms;
```

```
if (!formIsValid) {

    console.warn(
        "Booking validation failed."
    );



    /*
        Scroll to first error.
    */

    const firstError =
        document.querySelector(
            ".input-error"
        );



    if (firstError) {

        firstError.scrollIntoView({
            behavior: "smooth",
            block: "center"
        });

        firstError.focus();
    }



    return;
}



/*
    P4 - TRY/CATCH
    Registration processing.
*/

try {

    const registrationId =
        generateRegistrationId();



    bookingCount++;



    const {
        ticketName,
        ticketPrice
    } = getTicketInformation();



    const ticketQuantity =
        Number(quantity.value);



    const addons =
        getSelectedAddons();
```

```
const addonTotal =
    calculateAddonTotal(addons);


const subtotal =
    ticketPrice *
    ticketQuantity +
    addonTotal;


const discountAmount =
    Math.round(
        subtotal *
        discount /
        100
    );


const total =
    subtotal -
    discountAmount;


/*
    P6 - extract email domain
*/

const emailDomain =
    extractEmailDomain(
        email.value
    );


/*
    P6 - check student domain
*/

const studentEmail =
    isStudentEmail(
        email.value
    );


const booking = {

    registrationId,

    name:
        fullName.value.trim(),

    email:
        email.value.trim(),

    phone:
        phone.value.trim(),
```

```
    studentId:
        studentId.value.trim(),

    college:
        college.value.trim(),

    ticket:
        ticketName,

    quantity:
        ticketQuantity,

    session:
        session.options[
            session.selectedIndex
        ].text,

    addons,

    food:
        foodPreference.value,

    subtotal,

    discount:
        discountAmount,

    total,

    emailDomain,

    studentEmail,

    date:
        workshopDate
};



/*
    P9 - Save final booking
*/

sessionStorage.setItem(
    "latestBooking",
    JSON.stringify(booking)
);



localStorage.setItem(
    "lastRegistrant",
    fullName.value.trim()
);



console.table(booking);



showConfirmation(
```

```
                booking
            );


        } catch (error) {

            console.error(
                "Booking failed:",
                error
            );


            alert(
                "Something went wrong while processing your booking."
            );
        }
    }
);



/* ================================================
   P8 - RESET EVENT
================================================ */

bookingForm.addEventListener(
    "reset",
    function () {

        /*
            reset event fires before browser
            finishes resetting fields.
        */

        setTimeout(
            function () {

                attendees = [];

                discount = 0;

                appliedPromo = "";


                promoMessage.textContent = "";

                promoMessage.className =
                    "promo-message";


                document
                    .getElementById(
                        "discountRow"
                    )
                    .classList.add(
                        "hidden"
                    );
```

```
    renderAttendees();


    document
        .querySelectorAll(
            ".input-error, .input-success"
        )
        .forEach(
            element => {

                element.classList.remove(
                    "input-error",
                    "input-success"
                );
            }
        );


    document
        .querySelectorAll(
            ".error-message"
        )
        .forEach(
            error => {

                error.textContent = "";
            }
        );


    sessionStorage.removeItem(
        "workshopBooking"
    );


    sessionStorage.removeItem(
        "ticketQuantity"
    );


    sessionStorage.removeItem(
        "preferredSession"
    );


    sessionStorage.removeItem(
        "foodPreference"
    );


    updateSummary();


    console.info(
        "Booking form has been reset."
    );

},
0
```

```
        );
    }
);


/* ==================================================
   SHOW CONFIRMATION
================================================== */

function showConfirmation(booking) {

    /*
        P2 - Destructuring
    */

    const {
        registrationId,
        name,
        ticket,
        total,
        session: selectedSession
    } = booking;



    document.getElementById(
        "registrationId"
    ).textContent =
        registrationId;



    document.getElementById(
        "confirmationMessage"
    ).textContent =
        `Thank you ${name}! Your ${ticket} has been successfully booked.`;



    const confirmationDetails =
        document.getElementById(
            "confirmationDetails"
        );



    confirmationDetails.innerHTML = `

        <div>
            📅 ${booking.date}
        </div>

        <div>
            🎟️ ${ticket} × ${booking.quantity}
        </div>

        <div>
            💺 ${selectedSession}
        </div>

        <div>
            💰 Total Paid: ₹${total}
        </div>
```

```
    `;


    document
        .getElementById(
            "bookingStatus"
        )
        .textContent =
        "Confirmed";



    document
        .getElementById(
            "bookingStatus"
        )
        .style.background =
        "rgba(21, 153, 87, 0.12)";



    document
        .getElementById(
            "bookingStatus"
        )
        .style.color =
        "#159957";



    confirmationModal.classList.remove(
        "hidden"
    );



    console.info(
        `Booking confirmed: ${registrationId}`
    );
}



/* ====================================================
   CLOSE MODAL
==================================================== */

function closeConfirmation() {

    confirmationModal.classList.add(
        "hidden"
    );
}



closeModal.addEventListener(
    "click",
    closeConfirmation
);



doneButton.addEventListener(
    "click",
    closeConfirmation
```

```
);


/* ====================================================
   MODAL CLICK OUTSIDE
==================================================== */

confirmationModal.addEventListener(
    "click",
    function (event) {

        if (
            event.target ===
            confirmationModal
        ) {

            closeConfirmation();
        }
    }
);



/* ====================================================
   P1 - DYNAMIC WELCOME MESSAGE
==================================================== */

fullName.addEventListener(
    "input",
    function () {

        const name =
            fullName.value.trim();



        if (name) {

            console.log(
                `Welcome ${name}! You're booking a TechNova workshop ticket.`
            );
        }
    }
);



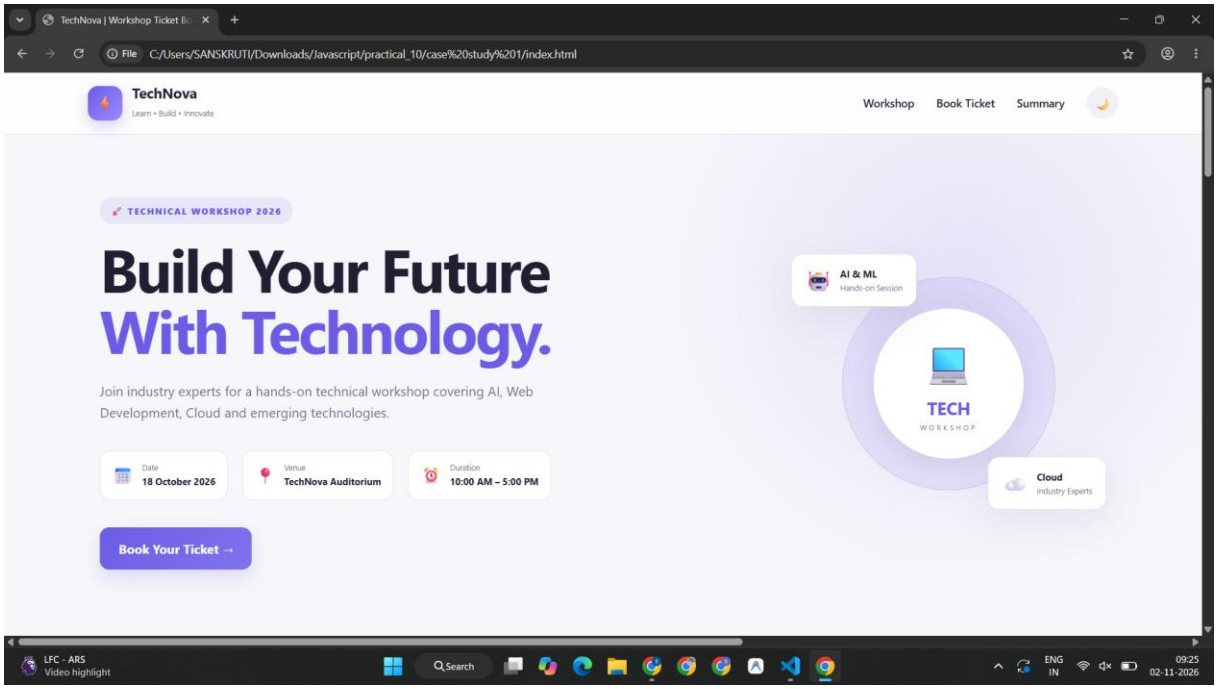
/* ====================================================
   INITIAL SUMMARY
==================================================== */

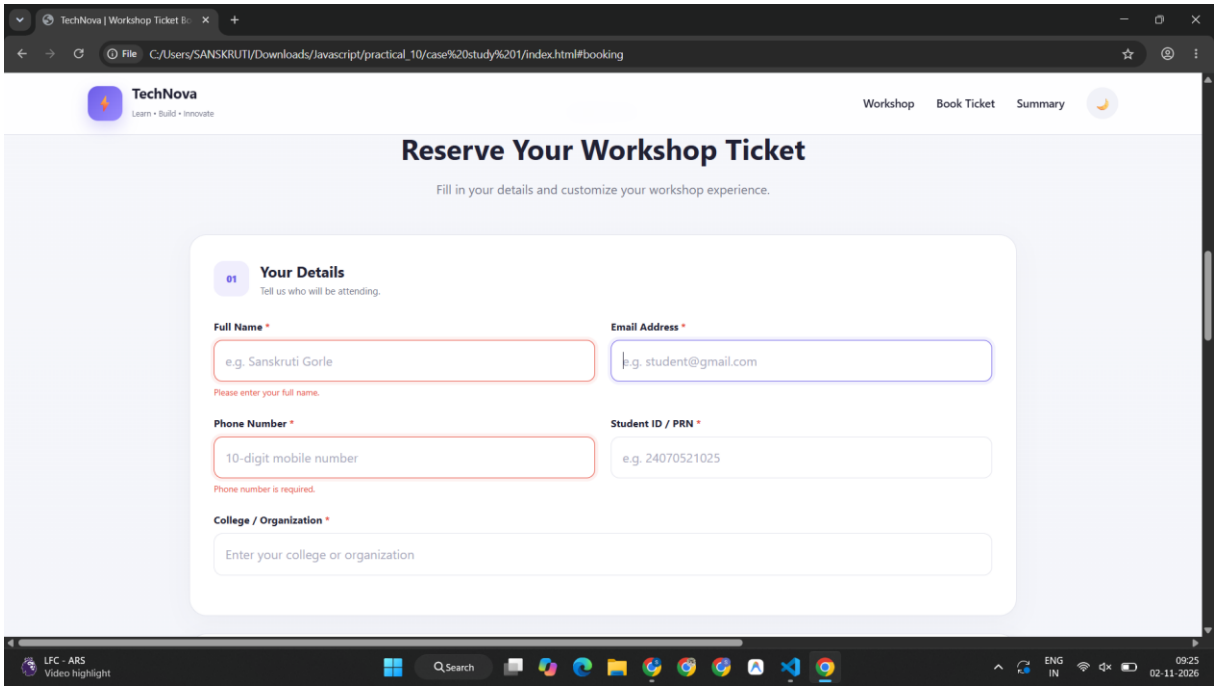
updateSummary();

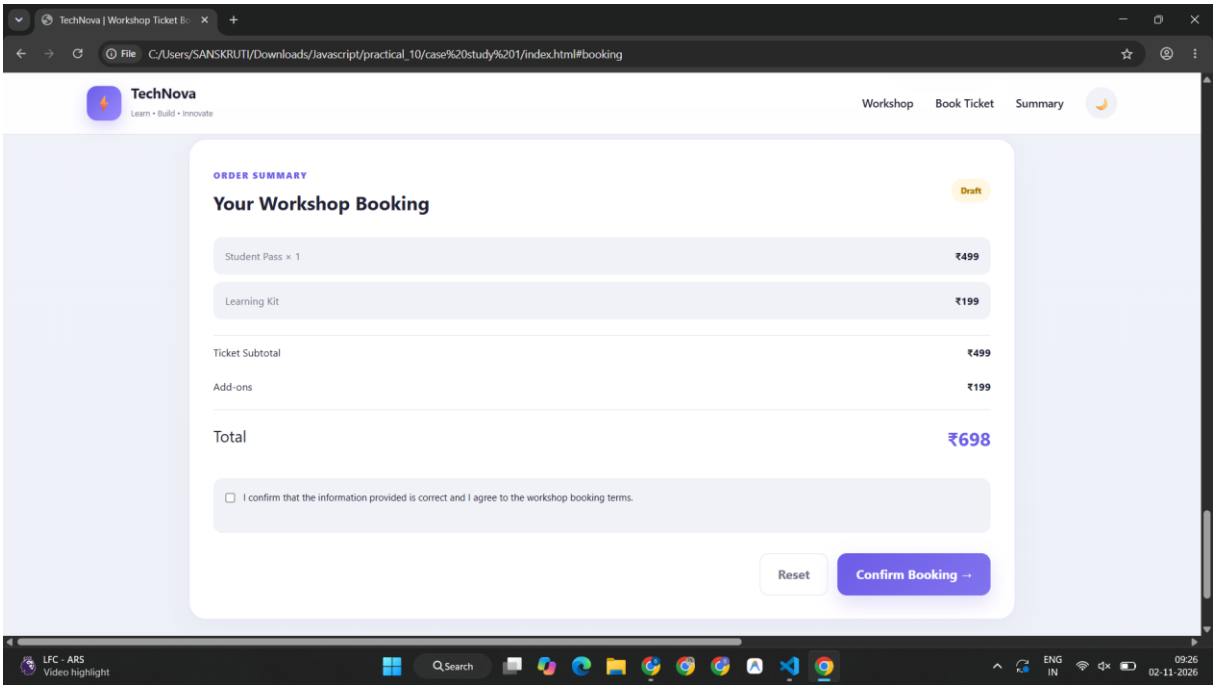


console.log(
    "Workshop booking application ready."
);
```

Output –


TechNova | Workshop Ticket Bo
File C:/Users/SANSKRUTI/Downloads/Javascript/practical_10/case%20study%201/index.html
TechNova
Learn • Build • Innovate
Workshop
Book Ticket
Summary
TECHNICAL WORKSHOP 2026
Build Your Future
With Technology.
Join industry experts for a hands-on technical workshop covering AI, Web Development, Cloud and emerging technologies.
Date
18 October 2026
Venue
TechNova Auditorium
Duration
10:00 AM – 5:00 PM
Book Your Ticket →
AI & ML
Hands-on Session
TECH
WORKSHOP
Cloud
Industry Experts
LFC - ARS
Video highlight
Search
ENG
IN
09:25
02-11-2026


TechNova | Workshop Ticket Bo
File C:/Users/SANSKRUTI/Downloads/Javascript/practical_10/case%20study%201/index.html#booking
TechNova
Learn • Build • Innovate
Workshop
Book Ticket
Summary
Reserve Your Workshop Ticket
Fill in your details and customize your workshop experience.
01
Your Details
Tell us who will be attending.
Full Name *
e.g. Sanskruti Gorle
Please enter your full name.
Email Address *
e.g. student@gmail.com
Phone Number *
10-digit mobile number
Phone number is required.
Student ID / PRN *
e.g. 24070521025
College / Organization *
Enter your college or organization
LFC - ARS
Video highlight
Search
ENG
IN
09:25
02-11-2026



Additional Module 1: Smart India Hackathon (SIH) Information Portal

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Smart India Hackathon</title>

    <link rel="stylesheet" href="style.css">
</head>
<body>

    <!-- Header -->
    <header>
        <h1>SMART INDIA HACKATHON</h1>
        <p>Innovate | Create | Solve</p>
    </header>

    <!-- Navigation Bar -->
    <nav>
        <a href="#home">Home</a>
        <a href="#about">About</a>
        <a href="#achievements">Achievements</a>
        <a href="#opportunities">Opportunities</a>
    </nav>

    <!-- Home Section -->
    <section id="home">
        <h2>Welcome to Smart India Hackathon</h2>

        <p>
            Smart India Hackathon (SIH) is a nationwide initiative
```

```
        that encourages students to develop innovative solutions
        to real-world problems.
    </p>

    <p>
        It promotes <b>innovation</b>, <i>creativity</i> and
        <u>teamwork</u> among students.
    </p>

    <button id="welcomeBtn">Click for Welcome Message</button>
</section>

<hr>

<!-- About Section -->
<section id="about">
    <h2>About SIH</h2>

    <p>
        Smart India Hackathon provides students with a platform
        to solve challenges presented by organizations and
        government departments.
    </p>

    <h3>Major Objectives</h3>

    <ul>
        <li>Encourage innovative thinking.</li>
        <li>Develop technical and programming skills.</li>
        <li>Promote teamwork and collaboration.</li>
        <li>Solve real-world challenges.</li>
    </ul>
</section>

<hr>

<!-- Achievements Section -->
<section id="achievements">
    <h2>Achievements and Benefits</h2>

    <p>Participating in hackathons helps students to:</p>

    <ul>
        <li>Gain practical project experience.</li>
        <li>Improve problem-solving abilities.</li>
        <li>Build technical portfolios.</li>
        <li>Collaborate with students from different institutions.</li>
    </ul>

    <p class="highlight">
        Learn new skills, share ideas and turn innovative concepts
        into practical solutions!
    </p>
</section>

<hr>

<!-- Opportunities Section -->
<section id="opportunities">
    <h2>Career Opportunities</h2>
```

```
        <p>
            Hackathon experience can help students develop skills
            useful for internships, projects and future employment.
        </p>

        <p>
            <b>Important Skills:</b><br>
            Programming<br>
            Problem Solving<br>
            Teamwork<br>
            Innovation
        </p>
    </section>

    <!-- Footer -->
    <footer>
        <p>&copy; 2026 Smart India Hackathon Information Page</p>
        <p id="date"></p>
    </footer>

    <script src="script.js"></script>

</body>
</html>
```

style.css

```
* {
    box-sizing: border-box;
}

body {
    margin: 0;
    font-family: Arial, sans-serif;
    background-color: #f1f5fc;
    color: #333;
    line-height: 1.6;
}

/* Header */
header {
    background-color: #123b70;
    color: white;
    text-align: center;
    padding: 30px 15px;
}

header h1 {
    font-size: 32px;
    margin: 0;
}

header p {
    font-size: 18px;
    margin-bottom: 0;
}

/* Navigation */
nav {
    background-color: #f28c28;
    text-align: center;
```

```
    padding: 14px;
}

nav a {
    color: white;
    text-decoration: none;
    font-weight: bold;
    margin: 0 15px;
}

nav a:hover {
    color: #123b70;
    text-decoration: underline;
}

/* Content Sections */
section {
    max-width: 900px;
    margin: 22px auto;
    padding: 25px;
    background-color: white;
    border-radius: 8px;
    box-shadow: 0 3px 10px rgba(0, 0, 0, 0.06);
}

h2 {
    color: #123b70;
    border-bottom: 2px solid #f28c28;
    padding-bottom: 8px;
}

h3 {
    color: #d66b12;
}

p {
    text-align: justify;
}

li {
    margin-bottom: 8px;
}

hr {
    max-width: 900px;
    border: none;
    height: 2px;
    background-color: #d5deeb;
}

.highlight {
    color: #d66b12;
    font-weight: bold;
}

/* Button */
button {
    background-color: #123b70;
    color: white;
    padding: 12px 18px;
    border: none;
```

```
    border-radius: 5px;
    cursor: pointer;
}

button:hover {
    background-color: #f28c28;
}

/* Footer */
footer {
    background-color: #123b70;
    color: white;
    text-align: center;
    padding: 15px;
    margin-top: 25px;
}

footer p {
    text-align: center;
    margin: 5px;
}

/* Responsive Design */
@media (max-width: 600px) {
    header h1 {
        font-size: 24px;
    }

    nav a {
        display: inline-block;
        margin: 5px 8px;
    }

    section {
        margin: 12px;
        padding: 18px;
    }

    hr {
        margin: 15px;
    }
}
```

script.js

```
// Welcome message button
const welcomeBtn = document.getElementById("welcomeBtn");

welcomeBtn.addEventListener("click", function () {
    alert("Welcome to Smart India Hackathon! Keep Innovating!");
});

// Navigation link interaction
const navLinks = document.querySelectorAll("nav a");

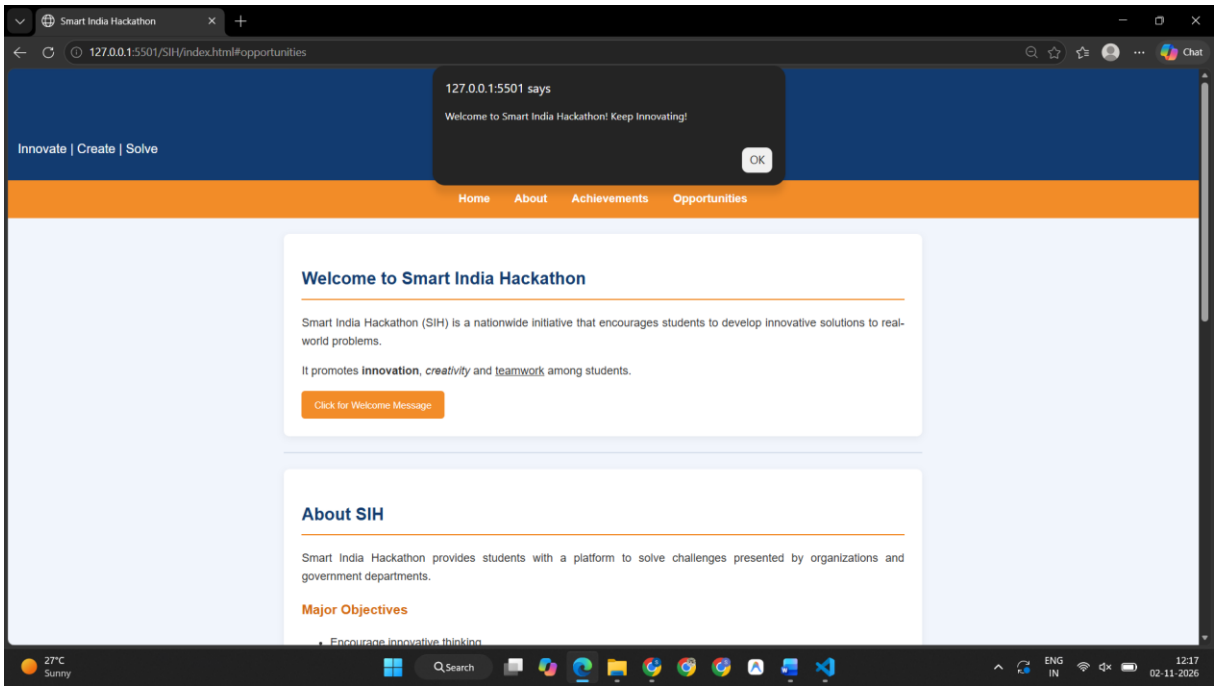
navLinks.forEach(function (link) {
    link.addEventListener("click", function () {
        console.log("Selected section: " + link.textContent);
    });
});
```

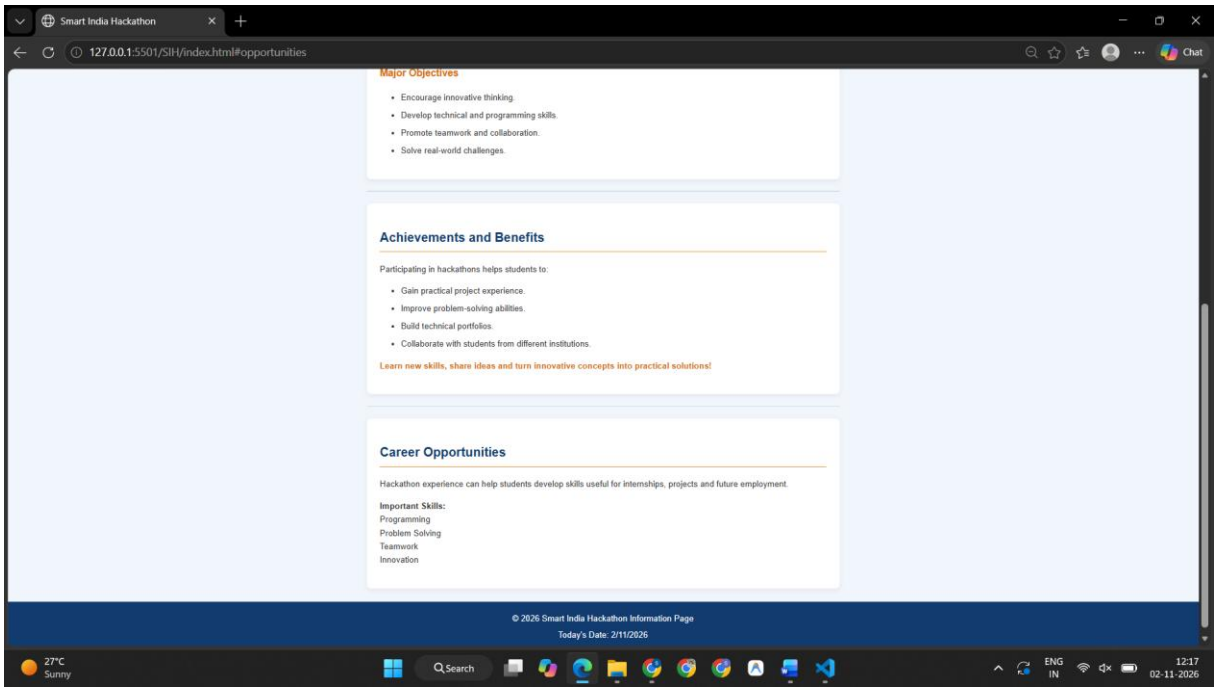
```
// Display current date in the footer
const dateElement = document.getElementById("date");

const today = new Date();

dateElement.textContent =
    "Today's Date: " + today.toLocaleDateString();
```

Output –





Additional Module 2: Rakhi World E-Commerce Store & Interactive Cart

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Rakhi World | Celebrate the Bond</title>

    <link rel="stylesheet" href="style.css">
</head>
<body>

    <!-- Header -->
    <header class="header">
        <div class="logo">Rakhi<span>World</span></div>

        <p>Celebrating the Bond of Love 💖</p>

        <div class="cart">
            🛍️ Cart: <span id="cartCount">0</span>
        </div>
    </header>

    <!-- Navigation -->
    <nav>
        <a href="#home">Home</a>
        <a href="#products">Shop Rakhis</a>
        <a href="#about">About Us</a>
        <a href="#contact">Contact</a>
    </nav>

    <!-- Hero Section -->
    <section id="home" class="hero">
        <div class="hero-content">
            <p class="tagline">A FESTIVAL OF LOVE</p>

            <h1>Make Every Bond<br>More Beautiful</h1>

            <p>
                Discover beautiful rakhis that express love,
                care and the special bond between siblings.
            </p>

            <a href="#products" class="shop-btn">Explore Collection →</a>
        </div>

        <div class="hero-decoration">
            🎁
            <span>Special Gifts, Special Bonds</span>
        </div>
    </section>

    <!-- Product Section -->
    <section id="products" class="products-section">

        <div class="section-heading">
            <p class="tagline">HANDPICKED WITH LOVE</p>
            <h2>Our Rakhi Collection</h2>
```

```
    <p>Find the perfect rakhi for your favourite person.</p>
</div>

<div class="product-container">

    <!-- Floral Rakhi -->
    <div class="product">
        <div class="product-image">
            <img src="./assets/Floral Rakhi with Kumkum Accents.png"
                 alt="Pink floral rakhi">
            <span class="badge">POPULAR</span>
        </div>

        <div class="product-info">
            <p class="category">FLORAL COLLECTION</p>
            <h3>Floral Rakhi</h3>
            <p class="description">
                A delicate floral design with beautiful pink petals.
            </p>

            <div class="price">₹99</div>

            <button class="buy-btn">Add to Cart</button>
        </div>
    </div>

    <!-- Designer Rakhi -->
    <div class="product">
        <div class="product-image">
            <img src="./assets/Ornate Red Gemstone Rakhi on Silk.png"
                 alt="Blue gemstone designer rakhi">
            <span class="badge">PREMIUM</span>
        </div>

        <div class="product-info">
            <p class="category">DESIGNER COLLECTION</p>
            <h3>Designer Rakhi</h3>
            <p class="description">
                An elegant blue gemstone rakhi with golden detailing.
            </p>

            <div class="price">₹149</div>

            <button class="buy-btn">Add to Cart</button>
        </div>
    </div>

    <!-- Traditional Rakhi -->
    <div class="product">
        <div class="product-image">
            <img src="./assets/Traditional Red and Gold Rakhi Still Life.png"
                 alt="Traditional red and gold Om rakhi">
            <span class="badge">TRADITIONAL</span>
        </div>

        <div class="product-info">
            <p class="category">CLASSIC COLLECTION</p>
            <h3>Traditional Rakhi</h3>
            <p class="description">
                A traditional red and gold rakhi with an Om design.
            </p>
```

```
                <div class="price">₹79</div>

                <button class="buy-btn">Add to Cart</button>
            </div>
        </div>

        <!-- Premium Rakhi -->
        <div class="product">
            <div class="product-image">
                <img src="./assets/Elegant Blue Gemstone Rakhi Setup.png"
                     alt="Premium Ganesha rakhi">
                <span class="badge">EXCLUSIVE</span>
            </div>

            <div class="product-info">
                <p class="category">LUXURY COLLECTION</p>
                <h3>Premium Rakhi</h3>
                <p class="description">
                    A beautiful Ganesha rakhi with rich red and gold details.
                </p>

                <div class="price">₹199</div>

                <button class="buy-btn">Add to Cart</button>
            </div>
        </div>

    </div>
</section>

<!-- Features -->
<section class="features">
    <div>
        <span>🎀</span>
        <h3>Beautiful Designs</h3>
        <p>Special designs for every sibling.</p>
    </div>

    <div>
        <span>🎁</span>
        <h3>Gift with Love</h3>
        <p>Make your celebration memorable.</p>
    </div>

    <div>
        <span>💝</span>
        <h3>Made for Family</h3>
        <p>Celebrate your unbreakable bond.</p>
    </div>
</section>

<!-- About -->
<section id="about" class="info-section">
    <h2>About Rakhi World</h2>

    <p>
        Rakhi World brings you a carefully selected collection
        of traditional, floral, designer and premium rakhis.
        Our aim is to make every Raksha Bandhan special through
```

```
            beautiful designs and meaningful gifts.
        </p>
    </section>

    <!-- Contact -->
    <section id="contact" class="info-section contact">
        <h2>Contact Us</h2>
        <p>Email: support@rakhiworld.example</p>
        <p>Phone: +91 98765 43210</p>
    </section>

    <!-- Footer -->
    <footer>
        <h2>Rakhi World</h2>
        <p>Celebrating love, one rakhi at a time. 💚</p>
        <p>© <span id="year"></span> Rakhi World. All rights reserved.</p>
    </footer>

    <script src="script.js"></script>

</body>
</html>
```

style.css

```
* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

html {
    scroll-behavior: smooth;
}

body {
    font-family: Arial, sans-serif;
    background: #fffaf7;
    color: #3e2928;
    line-height: 1.6;
}

/* Header */
.header {
    min-height: 100px;
    padding: 18px 5%;
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 20px;
    flex-wrap: wrap;
    background: white;
}

.logo {
    font-size: 30px;
    font-weight: bold;
    color: #a83250;
}

.logo span {
    color: #d59b47;
```

```
}

.header p {
    color: #765957;
}

.cart {
    padding: 10px 16px;
    border: 1px solid #ead4d5;
    border-radius: 25px;
    font-weight: bold;
}

/* Navigation */
nav {
    background: #7f203a;
    padding: 15px 5%;
    display: flex;
    justify-content: center;
    gap: 45px;
    flex-wrap: wrap;
}

nav a {
    color: white;
    text-decoration: none;
    font-weight: bold;
}

nav a:hover {
    color: #ffd6a5;
}

/* Hero Section */
.hero {
    min-height: 380px;
    padding: 65px 8%;
    display: flex;
    justify-content: space-between;
    align-items: center;
    gap: 40px;
    background: linear-gradient(120deg, #ffe8e8, #fff1dc);
}

.hero-content {
    max-width: 650px;
}

.tagline {
    color: #a83250;
    font-size: 12px;
    font-weight: bold;
    letter-spacing: 2px;
}

.hero h1 {
    font-size: clamp(32px, 4vw, 52px);
    line-height: 1.2;
    margin: 15px 0;
    color: #7f203a;
}
```

```
.hero-content > p:not(.tagline) {
    max-width: 520px;
    margin-bottom: 25px;
    color: #715553;
}

.shop-btn {
    display: inline-block;
    background: #a83250;
    color: white;
    padding: 13px 24px;
    border-radius: 5px;
    text-decoration: none;
    font-weight: bold;
}

.shop-btn:hover {
    background: #7f203a;
}

.hero-decoration {
    width: 280px;
    height: 220px;
    flex-shrink: 0;
    border-radius: 50%;
    background: rgba(255, 255, 255, 0.65);
    display: flex;
    flex-direction: column;
    justify-content: center;
    align-items: center;
    font-size: 85px;
    color: #a83250;
}

.hero-decoration span {
    font-size: 13px;
    font-weight: bold;
    text-align: center;
}

/* Products */
.products-section {
    padding: 65px 5%;
}

.section-heading {
    text-align: center;
    margin-bottom: 40px;
}

.section-heading h2 {
    color: #7f203a;
    font-size: 34px;
    margin: 8px 0;
}

.section-heading > p:last-child {
    color: #806b68;
}
```

```
/* Wide product layout */
.product-container {
    width: 100%;
    display: grid;
    grid-template-columns: repeat(4, minmax(0, 1fr));
    gap: 24px;
}

/* Product Cards */
.product {
    min-width: 0;
    overflow: hidden;
    background: white;
    border: 1px solid #f0dfe0;
    border-radius: 10px;
    transition: transform 0.25s, box-shadow 0.25s;
}

.product:hover {
    transform: translateY(-5px);
    box-shadow: 0 10px 25px rgba(100, 30, 50, 0.12);
}

.product-image {
    position: relative;
    height: 250px;
    background: #f9eee8;
    overflow: hidden;
}

.product-image img {
    width: 100%;
    height: 100%;
    object-fit: cover;
    display: block;
    transition: transform 0.35s;
}

.product:hover .product-image img {
    transform: scale(1.04);
}

.badge {
    position: absolute;
    top: 12px;
    left: 12px;
    padding: 5px 9px;
    border-radius: 4px;
    background: #a83250;
    color: white;
    font-size: 10px;
    font-weight: bold;
}

.product-info {
    padding: 20px;
}

.category {
    font-size: 10px;
    letter-spacing: 1px;
    color: #a83250;
```

```
    font-weight: bold;
}

.product-info h3 {
    margin: 7px 0;
    font-size: 20px;
    color: #452b2d;
}

.description {
    min-height: 65px;
    font-size: 13px;
    color: #806b68;
}

.price {
    margin: 12px 0;
    font-size: 23px;
    font-weight: bold;
    color: #a83250;
}

.buy-btn {
    width: 100%;
    padding: 12px;
    background: #a83250;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
    font-size: 14px;
    font-weight: bold;
}

.buy-btn:hover {
    background: #7f203a;
}

/* Features */
.features {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 25px;
    padding: 40px 8%;
    background: #fff0ec;
    text-align: center;
}

.features span {
    font-size: 35px;
}

.features h3 {
    margin: 8px 0;
    color: #7f203a;
}

.features p {
    color: #806b68;
    font-size: 14px;
}
```

```
/* About and Contact */
.info-section {
    max-width: 1000px;
    margin: 0 auto;
    padding: 45px 30px;
    text-align: center;
}

.info-section h2 {
    color: #7f203a;
    margin-bottom: 15px;
}

.info-section p {
    color: #715e5c;
}

.contact {
    padding-top: 15px;
}

/* Footer */
footer {
    background: #7f203a;
    color: white;
    text-align: center;
    padding: 28px 15px;
}

footer h2 {
    margin-bottom: 5px;
}

footer p {
    margin: 5px 0;
    font-size: 14px;
}

/* Responsive Design */
@media (max-width: 1100px) {
    .product-container {
        grid-template-columns: repeat(2, minmax(0, 1fr));
    }

    .product-image {
        height: 280px;
    }

    .hero-decoration {
        width: 220px;
        height: 190px;
    }
}

@media (max-width: 600px) {
    .header {
        justify-content: center;
        text-align: center;
        flex-direction: column;
    }
```

```
    nav {
        gap: 15px;
        padding: 15px 10px;
    }

    nav a {
        font-size: 13px;
    }

    .hero {
        padding: 45px 22px;
    }

    .hero-decoration {
        display: none;
    }

    .hero h1 {
        font-size: 34px;
    }

    .products-section {
        padding: 40px 15px;
    }

    .section-heading h2 {
        font-size: 27px;
    }

    .product-container {
        grid-template-columns: 1fr;
        gap: 20px;
    }

    .product-image {
        height: 300px;
    }

    .features {
        grid-template-columns: 1fr;
    }
}
```

script.js

```
// Select all product buttons
const buyButtons = document.querySelectorAll(".buy-btn");

let cartCount = 0;

// Add event to every product button
buyButtons.forEach(function (button) {

    button.addEventListener("click", function () {

        const product = button.closest(".product");

        const productName =
            product.querySelector("h3").textContent;
```

```
        const productPrice =
            product.querySelector(".price").textContent;

        cartCount++;

        document.getElementById("cartCount").textContent =
            cartCount;

        alert(
            productName + " added to your cart!\n" +
            "Price: " + productPrice + "\n" +
            "Items in cart: " + cartCount
        );

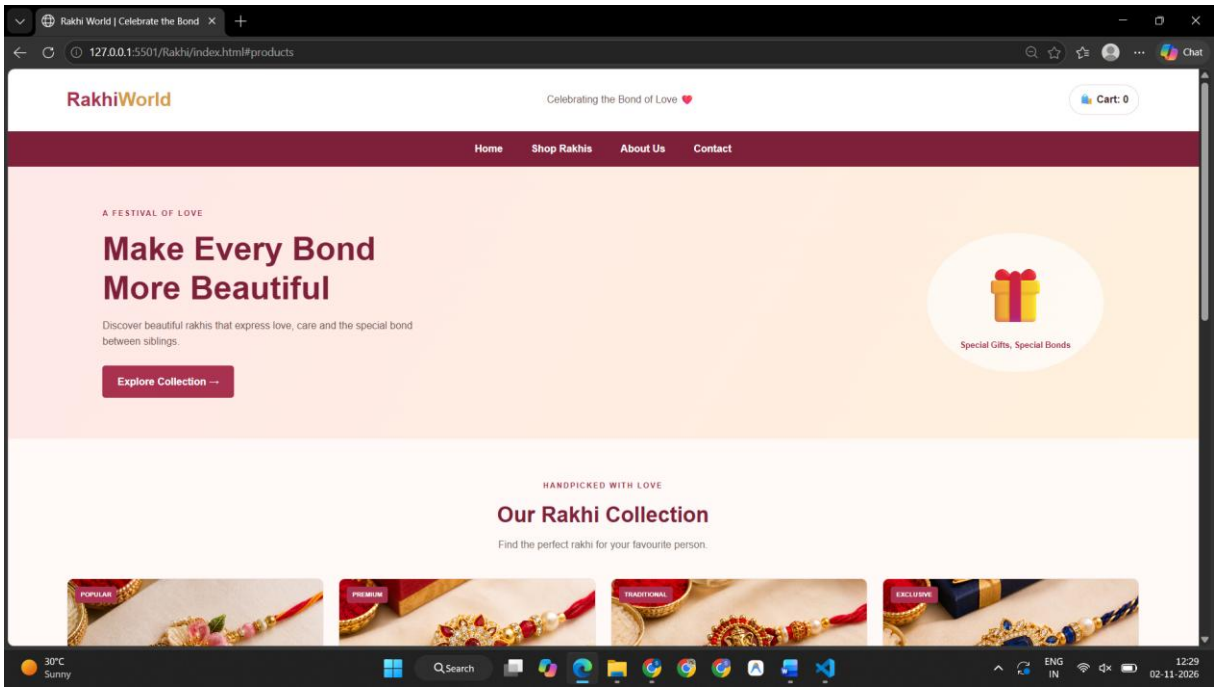
    });

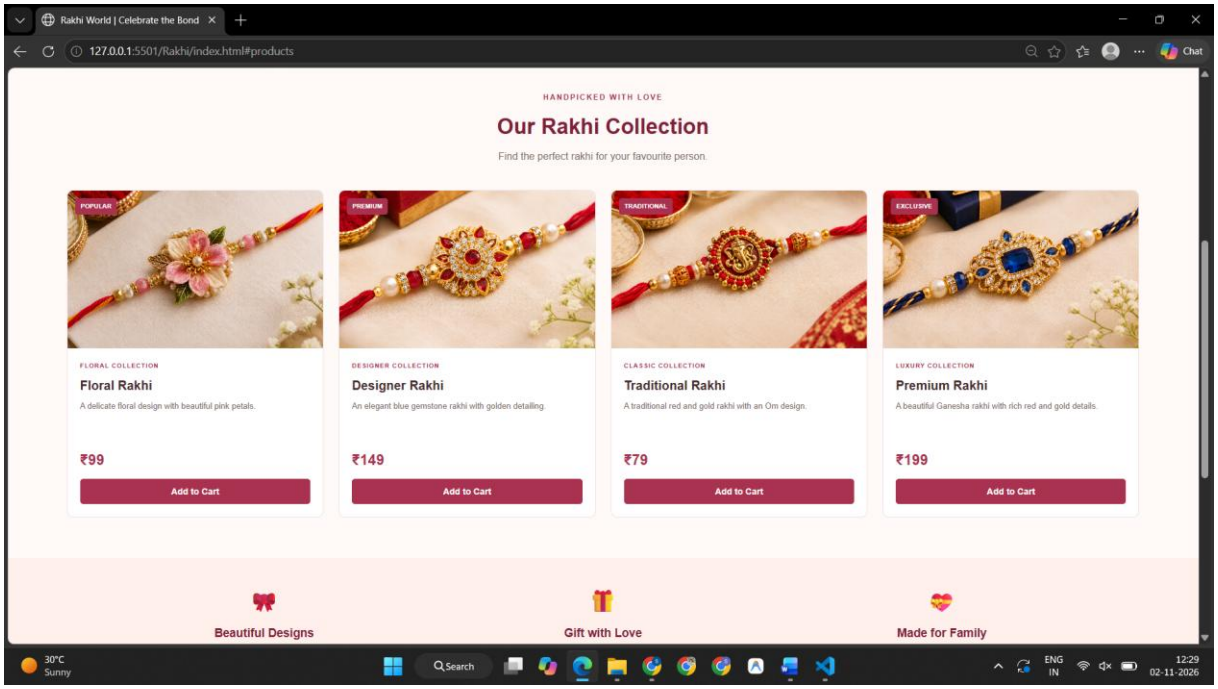
});

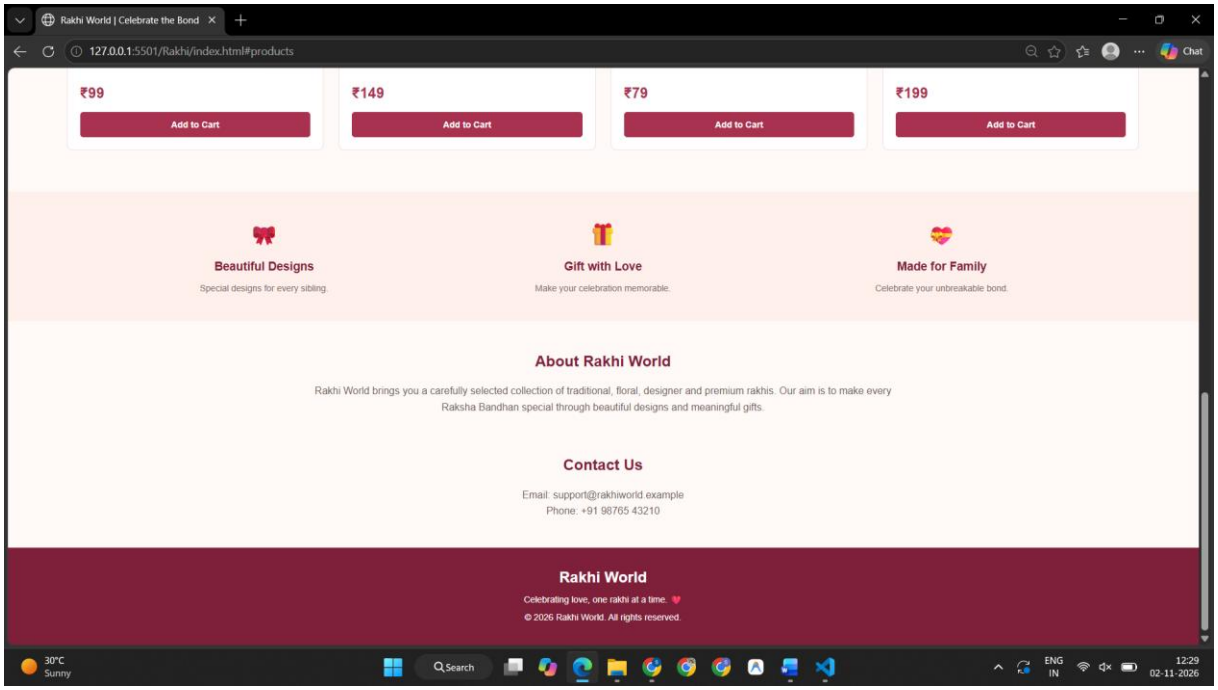
// Display current year automatically
document.getElementById("year").textContent =
    new Date().getFullYear();

// Log a message in the console
console.log("Welcome to Rakhi World!");
```

Output –







Conclusion –

In this experiment, JavaScript timing functions (setInterval, setTimeout, clearInterval) and ES6 modular programming (import/export) were thoroughly understood and implemented. An interactive Countdown Timer and Class Scheduler was successfully created with dynamic state updates. Furthermore, Case Study 1 (TechNova Workshop Booking) demonstrated comprehensive ticket calculation, coupon code application, live DOM feedback, and form validation. The Smart India Hackathon (SIH) portal and Rakhi World e-commerce showcase further highlighted interactive event handling, dialog alerts, and dynamic product cart tracking. All applications were tested and verified successfully in the browser.